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Journal of Manufacturing Processes

Date: 30 September 2025

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# Arc instability in hybrid laser-GMAW welding process using Ni based filler wire NiCrMo-4

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10.1016/j.jmapro.2025.06.029 ↗

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## Abstract

NiCrMo-4 alloy, due to its excellent low-temperature properties and low ductile-to-brittle transition temperature, is widely used as a filler material for welding critical cryogenic structures such as large LNG storage tanks. However, the ER-NiCrMo-4 wire, with its high melting point, viscosity, and surface tension, tends to soften and exhibit irregular whipping at the wire tip during gas metal arc welding (GMAW) at currents far below the critical spray transfer current. This results in process instability and limits the application of GMAW in welding the inner tanks of LNG storage systems, where high safety and reliability are required. This study first investigates the instability mechanisms of ER-NiCrMo-4 wire during GMAW, and then examines the effects of laser assistance on droplet transfer behavior by analyzing the coupling between laser and arc plasma under different laser-wire distances. Results show that the laser can regulate the droplet

## Outline

Abstract

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1. Introduction

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CRediT authorship contribution statement

Declaration of competing interest

transfer stability of ER-NiCrMo-4 wire, and arc stability is influenced by the degree of coupling between the laser and arc plasmas, as well as the competing effects of metal vapor reaction forces. When the laser-to-arc axial distance ( $D_{LA}$ ) is set between 2 and 3 mm, a relatively stable arc is achieved.

Keywords

LNG storage tanks; Rotating spray transfer;  $D_{LA}$ ; Mechanics analysis; ER-NiCrMo-4

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List of symbols and acronyms

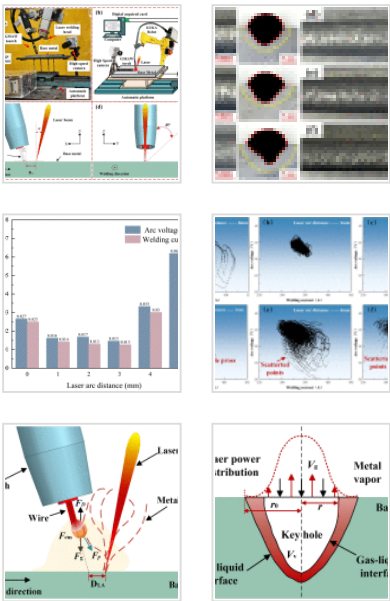
|                                                                   |                        |            |                                 |
|-------------------------------------------------------------------|------------------------|------------|---------------------------------|
| density of liquid metal $\rho_m$ (g/cm <sup>3</sup> )             | 8.89                   | $\gamma$   | surface tension coefficient     |
| atomic mass of the metal vapor $M_a$ (g/mol)                      | 58.69                  | $V_g$      | metal vapor jet velocity        |
| latent heat of evaporation $L_v$ (kJ/kg)                          | 340.7                  | $\rho_g$   | density of metal vapors         |
| Avogadro's number $N_a$ (mol <sup>-1</sup> )                      | $6.02 \times 10^{23}$  | rd         | radius of the molten droplet    |
| Boltzmann constant $K_b$ (J/K)                                    | $1.38 \times 10^{-23}$ | LAHW       | Laser-arc hybrid welding        |
| flow resistance coefficient $CD$                                  | 0.45                   | $D_{LA}$   | Distances between laser and arc |
| vaporization constant $BO$ (g·cm <sup>-1</sup> ·s <sup>-2</sup> ) | $3.9 \times 10^{13}$   | $C_v$      | Coefficient of variation        |
| Materials constants <b>A</b>                                      | 0.55                   | $\bar{U}$  | Average voltage                 |
| permeability of free space $\mu_0$ (Hm <sup>-1</sup> )            | $4\pi \times 10^{-7}$  | $\bar{I}$  | Average current                 |
| acceleration due to gravity $g$ (m/s <sup>2</sup> )               | 9.8                    | $\sigma_U$ | Standard deviation of voltage   |

Acknowledgments

References

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|                                                                                   |                   |            |                                    |
|-----------------------------------------------------------------------------------|-------------------|------------|------------------------------------|
| metal vapor drag coefficient <b>CP</b>                                            | 0.45              | $\sigma_I$ | Current standard deviation         |
| speed of sound in condensed phase <b>V<sub>0</sub></b> (m/s)                      | $3.4 \times 10^2$ | $F_D$      | Reactive force of the metal vapors |
| radius of the welding wire <b>r<sub>w</sub></b> (mm)                              | 0.6               | $F_g$      | Gravity                            |
| perfect gas constant <b>R<sub>0</sub></b> (J·mol <sup>-1</sup> ·K <sup>-1</sup> ) | 8.314             | $F_p$      | Plasma flow force                  |
| Vaporization potential <b>U</b> (K)                                               | 2406              | $F_s$      | Surface tension                    |
| surface temperature of the molten pool <b>Ts</b> (K)                              | 1900              | $F_{em}$   | Electromagnetic force              |

1. Introduction

NiCrMo-4 nickel-based alloy, known for its excellent high-temperature performance, low thermal expansion coefficient, and low ductile-to-brittle transition temperature, is widely used as a filler material in applications such as aerospace turbine engine components, large liquefied natural gas (LNG) storage tanks, and chemical pipeline welding [1]. With the global shift toward cleaner energy sources, LNG has gained increasing prominence in emission reduction and climate change mitigation, driving a significant rise in demand for its safe transportation and storage.

Large-scale LNG tanks, which operate under cryogenic conditions of -161.5°C and 0.1 MPa [2], serve as key infrastructure for efficient LNG storage and transport. Common materials for these tanks include austenitic stainless steels, Invar alloys, aluminum alloys, and 9% nickel steels. Among them, 9Ni steel is the primary structural material due to its high strength, excellent low-temperature toughness, and cost-effectiveness [3]. However, NiCrMo-4 alloy, as a filler metal, presents challenges during welding due to its poor flowability and high viscosity [4].

Currently, mainstream welding techniques for medium-thick 9Ni steel plates include submerged arc welding (SAW) [5], narrow-gap gas metal arc welding (GMAW) [6], and shielded metal arc welding (SMAW) [7]. With the increasing scale and standardization of

LNG tank fabrication, there is a pressing need for efficient and advanced welding technologies. Laser-Arc Hybrid Welding (LAHW), which integrates laser and arc heat sources to enable precise energy control [8], offers advantages such as high efficiency, low distortion, deep penetration, narrow heat-affected zones, adaptability to various materials and thicknesses, and automation compatibility [9]. These attributes make LAHW particularly suitable for LNG tank welding [10].

In LAHW, droplet transfer modes—short-circuiting, globular, and spray transfer—are influenced by process parameters. Spray transfer is particularly favorable for achieving stable processes and high-quality welds. Recent studies have focused on improving droplet behavior through approaches such as rotating arcs and laser modulation. Zhu et al. [11] introduced a hybrid process incorporating a rotating arc assisted by laser, achieving defect-free dual-pass welding of 21 mm thick 9Ni steel. Wei et al. [12] demonstrated that centrifugal forces from the rotating arc significantly improved molten pool flow and reduced fusion defects. Given the crucial role of droplet transfer stability in weld quality, it is essential to further investigate the underlying interactions between droplet behavior and heat sources to enhance welding stability.

In conventional gas metal arc welding (GMAW), the static force balance theory is often used to explain droplet transfer behavior, which results from the combined effects of surface tension, electromagnetic forces, plasma flow impingement, and gravity [13]. In LAHW, however, the interaction between the laser-induced and arc-induced plasmas alters the droplet's force equilibrium, making the process more complex [14]. Li et al. [15] showed that by using low-power lasers to control arc length, droplet transfer frequency could be increased and spatter suppressed. Liu et al. [16] highlighted the importance of droplet transfer stability in weld formation and spatter control, noting that the transfer frequency is affected by the interplay between electromagnetic forces and the recoil pressure from metal vapor. Sheng et al. [17] found that laser-assisted keyholes can attract and compress arc columns, shorten arc length, and suppress disturbances from metal vapor, thereby enhancing arc stability [18]. Studies have shown that the plasma coupling mechanism between the laser and arc in LAHW can be optimized by adjusting the laser

power and the laser-to-arc axial distance ( $D_{LA}$ ) [19]. However, in practical welding scenarios, laser power is often fixed due to groove geometry and gap constraints [20]. Thus, adjusting  $D_{LA}$  becomes a key strategy for effectively utilizing laser energy, particularly in the welding of nickel-based alloys, where it helps mitigate the adverse effects of laser plasma on droplet transfer.

Liu et al. [21] found that increasing  $D_{LA}$  promotes a transition in droplet transfer mode from globular to spray. Zhang et al. [22] reported, through spectral intensity analysis, that a larger  $D_{LA}$  weakens plasma coupling between the laser and arc, reduces forces acting on the droplet, and enhances surface tension-driven transfer, thereby stabilizing the process and reducing spatter. Cai et al. [23] further noted that at  $D_{LA}=0\text{mm}$ , composite plasma fluctuates violently with severe spatter and molten pool instability, whereas at  $D_{LA}=2\text{mm}$ , keyhole fluctuation is suppressed and pool stability is significantly improved.

In summary, the relationship between  $D_{LA}$  and welding stability has become a focal point in LAHW research. However, due to the high viscosity and poor flowability of C-276 nickel-based alloy, its droplet transfer mechanism under high-current welding conditions differs markedly from conventional materials [24]. Further investigation is needed to understand its stability and transfer behavior. In this study, a custom-developed current-voltage acquisition and analysis system is employed to evaluate welding process stability. High-speed imaging is used to observe droplet transfer and plasma dynamics under varying  $D_{LA}$  conditions. Combined with the static force balance theory, this work aims to clarify the influence of laser energy on droplet behavior and to reveal the regulatory mechanism of  $D_{LA}$  on droplet transfer stability. The findings will provide theoretical support for the optimization of LAHW processes for 9Ni steel and promote its application in the fabrication of inner LNG tank structures.

## 2. Materials and experimental procedure

This study used a 9Ni steel plate of dimensions  $300\text{mm}\times 100\text{mm}\times 20\text{mm}$  as the base material and a ER-NiCrMo-4(C-276) nickel-based alloy welding wire of diameter 1.2mm

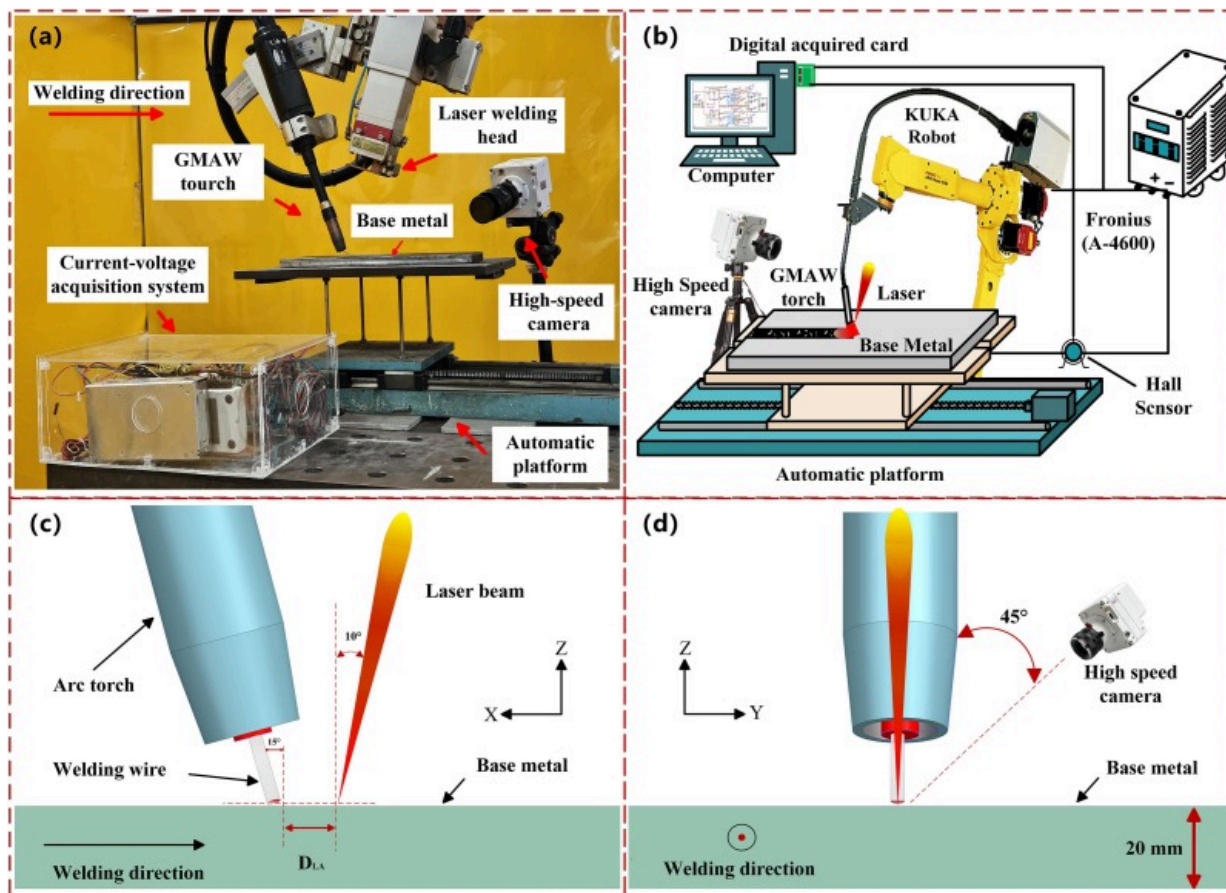
as the filler metal. [Table 1](#) shows the chemical composition of the base material and the welding wire. Before welding, the surface of the base material was polished and cleaned with acetone to remove surface contaminants, Pure argon gas was used as the shielding gas during the welding process.

Table 1. Chemical composition of the base material and welding wire (wt%).

|                    | Si     | Ti    | Mn    | Cu     | W     | Cr    | Ni      | Fe      | Mo    | Nb+Ta  | Al    |
|--------------------|--------|-------|-------|--------|-------|-------|---------|---------|-------|--------|-------|
| <b>9%Ni steel</b>  | 0.194  | 0.003 | 0.616 | <0.005 | <0.03 | 0.014 | 9.02    | Balance | 0.076 | <0.020 | 0.028 |
| <b>ER-NiCrMo-4</b> | <0.081 | –     | 0.43  | 0.45   | 0.34  | 15.78 | Balance | 6.08    | 16.31 | 0.34   | –     |

The welding experiment system ([Fig. 1](#)(a) and (b)) comprised an IPG YLS-6000 fiber laser, a KUKA robot, a Fronius A-4600 GMAW machine, and a computer numerical control (CNC) automatic platform. The IPG fiber laser heat source has a laser beam wavelength of 1.06μm, a focal length of 300mm, a maximum power of 4kW, a beam parameter product of 8mm·mrad, and a laser beam energy density with Gaussian distribution [\[27\]](#). The laser beam diameter at the focus point is 2mm. The laser device was tilted 10° in the vertical direction to avoid damage to the laser optical system due to laser reflection during the welding process [\[28,29\]](#). The Fronius (AR 5000) welding gun was inclined at an angle of 15° in the vertical direction. The laser device and GMAW welding torch were integrally assembled in a near-axis manner through a three-axis adjustable fixture developed by the research group. This allows for accurate fine-tuning of the relationship between the laser beam and the welding wire. The spatial position of the laser–arc composite head was controlled by a six-axis KUKA robot. The heat source layout used a laser-guided arc structure as shown in [Fig. 1](#)(c).





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Fig. 1. (a) Experimental platform. (b) Schematic of the experimental platform. (c) Schematic diagram showing the positions of the laser and the arc in the front view. (d) Schematic of the high-speed camera acquisition position.

For effective observation of the molten droplet transition process, welding process needs to remain stationary. This was achieved by the use of a self-made movable CNC automatic platform. The infrared temperature measurement system is the German DIAS shortwave infrared imager (PYROVIEW 512N) with a temperature measurement range of 900–

2500°C and a spectral span of 0.8–1.1  $\mu\text{m}$ . It features a high dynamic silicon detector array with 512× 378 pixels, enabling real-time sensing of the molten pool temperature throughout the welding process. A high-speed camera (Phantom VEO 340L) equipped with a Canon AF MICRO 200mm 1:4D lens was employed to observe the welding process from the side at a capture frequency of 3000 frames per second ([Fig. 1\(d\)](#)). The electrical signals from the welding process were collected using the NI 6211 data acquisition card in conjunction with a Hall sensor. The research group developed a LabVIEW data acquisition and analysis program to assess the stability of the current and voltage signals. Sampling frequency of the electrical signals is 3000 frames per second.

This study focuses on analyzing the impact of the  $D_{LA}$  on droplet transition stability in the C-276 wire LAHW process, specifically examining the droplet transition behavior for  $D_{LA}$  in the range 0–5 mm. All remaining welding process parameters for the LAHW remain constant, as shown in [Table 2](#). The selection of welding speed and laser power was based on my previous related research, using the experimental parameters that yielded the best stability.

Table 2. Parameters of LAHW.

| Parameters                                    | Value         |
|-----------------------------------------------|---------------|
| Current mode                                  | Standard mode |
| Laser power (kW)                              | 3             |
| Welding speed (m/min)                         | 0.55          |
| Defocus amount (mm) [ <a href="#">25,26</a> ] | −1            |
| Wire feeding rate (m/min)                     | 17            |
| Wire extension (mm)                           | 15            |
| Shielding gas flow rate (L/min)               | 25            |



| Parameters | Value |
|------------|-------|
| $M^2$      | <1.1  |

### 3. Results and discussion

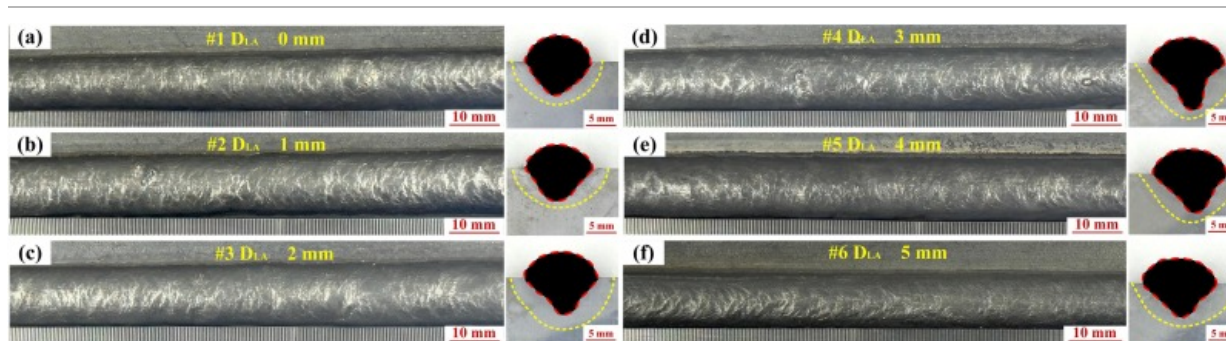
#### 3.1. Effects of laser–arc distance on arc stability

Based on previous research [[30], [31], [32]],  $D_{LA}$  is a key factor affecting the optimal coupling of laser and arc heat sources.  $D_{LA}$  significantly influences weld penetration, droplet transition mode, welding process stability, and the heat flow distribution of the heat sources.

As shown in Fig. 2, the macroscopic morphology and cross-sectional profiles of the welds under different laser-arc distances exhibit sound formation without the presence of typical metallurgical defects such as cracks or porosity. With the increase of the laser-arc distance from 0mm to 3mm, the weld penetration gradually increases, indicating an enhanced synergistic effect between the laser and the arc. In this range, the laser and arc interaction zones moderately overlap, allowing the laser beam to more readily sustain a stable keyhole due to the elevated thermal environment provided by the arc.

Simultaneously, the arc contributes to auxiliary melting around the keyhole region, resulting in effective energy coupling and complementary heat input from both heat sources. This synergistic interaction significantly improves the deep penetration capability of the molten pool. At a laser-arc distance of 3mm, the maximum weld penetration is achieved. However, when the distance further increases to 4–5mm, the weld penetration begins to decline, indicating a gradual reduction in the coupling effect between the laser and arc. As the spatial separation between the two heat sources increases, the laser and arc tend to act on distinct regions, failing to generate a unified molten pool. Consequently, the high energy density of the laser can no longer collaborate

effectively with the broad heat input of the arc, and the heat source complementarity is weakened.



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Fig. 2. The weld bead morphology and weld depth under different  $D_{LA}$ .

This study examined the impact of laser on the stability of the transfer process of nickel-based alloys by analyzing the coefficients of variation and cycle graphs of welding current and arc voltage at different  $D_{LA}$ . It found that the addition of laser can effectively enhance the stability of the transfer of high-viscosity metals. Additionally, the  $D_{LA}$  significantly affects the stability of the welding process. As the  $D_{LA}$  increases, the stability of the welding process initially improves and then deteriorates.

Data collected by current and voltage acquisition systems is crucial for evaluating the stability of the welding, energy input, and detecting any anomalies. Additionally, this data can indirectly reflect the physical properties of plasma, including conductivity, current density, and impedance. In the nickel-based alloys welding process, the transition is difficult due to their high viscosity and the large surface tension of melted wire tip droplets, which differs from materials such as low-carbon steel. Therefore, it is not sufficient to evaluate nickel-based alloys based solely on droplet transition frequency and critical droplet size. Moreover, there is limited discussion on the droplet transition process of nickel-based alloys. To evaluate the impact of varying  $D_{LA}$  on arc stability in

LAHW processes, this study characterizes the process through four aspects: average voltage ( $\bar{U}$ ), average current ( $\bar{I}$ ), standard deviation of voltage ( $\sigma_U$ ), and current standard deviation ( $\sigma_I$ ) [33].

Average voltage and current in LAHW are used to evaluate the overall level of arc current and voltage throughout the welding procedure. They help evaluate how laser energy affects arc energy at various  $D_{LA}$  settings. Their calculation formulas are as follows:

$$\bar{U} = \frac{\int_0^T U dt}{T} \quad (1)$$

$$\bar{I} = \frac{\int_0^T I dt}{T} \quad (2)$$

The standard deviations of current and voltage show the extent of variation between the actual current and voltage values throughout the welding process and their respective averages. They are instrumental in assessing the stability of the welding process. The mathematical expression for the current standard deviation is given in Eq. (3), and the mathematical expression for the voltage standard deviation is given in Eq. (4).

$$\sigma_U = \sqrt{\frac{1}{N} \sum_{i=1}^N (U_i - \bar{U})^2} \quad (3)$$

$$\sigma_I = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - \bar{I})^2} \quad (4)$$

In the formulas,  $\bar{U}$  represents the result of the calculation from Eq. (1), and  $\bar{I}$  represents the calculation result of Eq. (2).  $N$  denotes the total number of sampled data points,  $T$  signifies the total duration of the data points,  $U_i$  signifies the voltage value of each sampled point, and  $I_i$  signifies the current value of each sampled point.

This study calculated the average voltage and current, as well as the standard deviations of current and voltage, for cases without laser and with  $D_{LA}$  of 0, 1, 2, 3, 4, and 5 mm. The results of these calculations are presented in Table 3.

Table 3. Average current and voltage, and the standard deviation of current and voltage, at different  $D_{LA}$ .

| Test | $D_{LA}(\text{mm})$ | $\bar{U}(\text{V})$ | $\sigma_U$ | $\bar{I}(\text{A})$ | $\sigma_I$ |
|------|---------------------|---------------------|------------|---------------------|------------|
| 1    | Without laser       | 34.84               | 4.28       | 326.13              | 46.2       |
| 2    | 0                   | 33.03               | 0.88       | 344.53              | 8.61       |
| 3    | 1                   | 33.27               | 0.54       | 340.72              | 4.85       |
| 4    | 2                   | 33.15               | 0.56       | 339.72              | 4.43       |
| 5    | 3                   | 33.45               | 0.49       | 337.66              | 4.3        |
| 6    | 4                   | 33.63               | 1.12       | 334.98              | 10.09      |
| 7    | 5                   | 34.09               | 2.11       | 326.66              | 12.46      |

According to the data calculated in Table 3, when the laser is not applied, the average current is 326.13 A and the average voltage is 34.84V, and the standard deviations of current and voltage are 46.2 and 4.28, respectively. With the addition of the laser, the average voltage of the arc decreases overall, whereas the average current increases. The standard deviations of voltage ( $\sigma_U$ ) and current ( $\sigma_I$ ) decrease initially and then increase. The main reasons for this phenomenon are twofold. First, in the process of nickel-based alloy LAHW, the low-temperature laser contacts the high-temperature MIG arc, forming a large temperature gradient between the two. This gradient promotes energy and material exchange between them, resulting in an elevation in electron temperature within the hybrid zone [34]. Second, the metal on the molten pool's surface evaporates and ionizes in large quantities under the action of the laser, causing a rise in metal vapor density within the arc column zone. These metal vapors possess a lower ionization potential compared to the welding shielding gas. Under the influence of the metal vapors, the electron density in the arc column zone of the hybrid welding increases, forming a high-conductivity electrical channel that compresses the arc [35].

With the addition of the laser, there is minimal change in voltage as the  $D_{LA}$  varies between 0mm to 4mm. The parameter  $\sigma_1$  an initial decline followed by an increase, indicating that within a certain range of wire distance, the synergy between the laser and arc heat sources can be notably enhanced. When the  $D_{LA}$  is 5mm, the standard deviation values for voltage and current are 2.11 and 12.46, respectively. This is because the laser and arc are far apart at this distance and unable to provide heat at the same location. Additionally, the metal vapors generated by the laser beam acting on the molten pool are far from the arc, which makes forming a stable electrical channel difficult. Hence, the coupling efficiency between the laser beam and the arc is low [36].

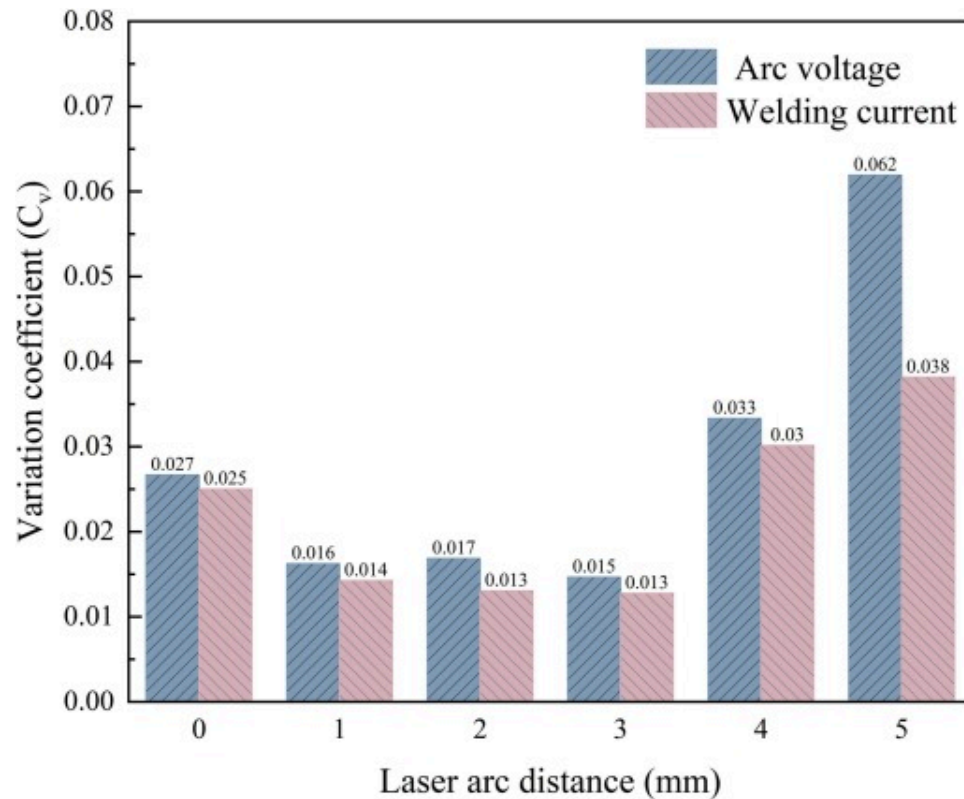
The results indicate that the addition of the laser can stabilize the welding process in the LAHW of nickel-based alloys, compared to GMAW welding. As the  $D_{LA}$  increases from 0mm to 5mm, the standard deviation values of current and voltage exhibit an initial decrease followed by an increase. When the wire distance was between 2mm and 3mm, there is an optimal coupling state between laser and arc energy, resulting in the most stable electrical signal.

To delve deeper into the impact of  $D_{LA}$  on arc stability, this study calculates the coefficient of variation (Cv) for current and voltage under different  $D_{LA}$  to gauge the dataset's dispersion [37,38]. The calculation formula is:

$$C_v = \frac{\sigma}{\mu} = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \mu)^2}}{\frac{1}{n} \sum_{i=1}^n x_i} \quad (5)$$

In the equation,  $\mu$  represents the average value of the gathered current and voltage dataset,  $\sigma$  represents the standard deviation of the collected current and voltage dataset,  $x_i$  represents the data value of the collected sample point, and  $n$  stands for the total number of data points. In the LAHW process, Cv can objectively characterize the fluctuation of the voltage and current parameters. The larger the value of Cv, the less stable the welding arc and welding process.

Fig. 3 shows the relationship between different  $D_{LA}$  and the  $C_v$  of welding current and arc voltage. When the  $D_{LA}$  is 0mm, the  $C_v$  is 0.025 for current and 0.027 for voltage. As the  $D_{LA}$  increases, the  $C_v$  of current and voltage initially decreases before rising again. At a  $D_{LA}$  of 3mm, the  $C_v$  for voltage and current reaches the minimum value, indicating that the arc stability is highest.



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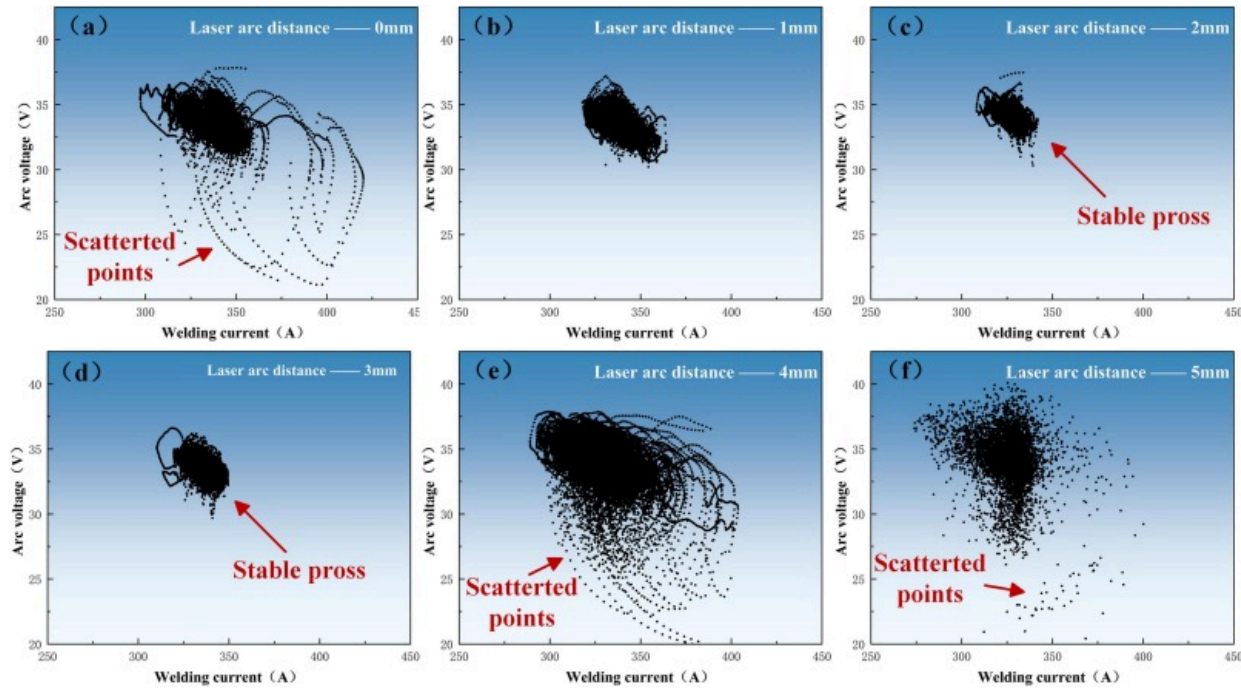
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Fig. 3. Variability coefficients ( $C_v$ ) for arc voltage and welding current.

Fig. 4 compares the electrical signals during the welding process at  $D_{LA}$  0mm to 5mm. When the  $D_{LA}$  is 0mm, unstable characteristic points are observed (Fig. 4(a)). The main cause of instability at this point is attributed to the vaporization of metal generated when



the laser interacts with the molten pool's surface, affecting the stability of droplet transition and resulting in significant fluctuations in the electrical signals [39]. When the  $D_{LA}$  is 1 mm, the characteristic points are relatively concentrated (Fig. 4(a)), indicating an improvement in arc stability. However, the range of fluctuation in current and voltage persists significantly during the welding process (Fig. 4(b)). This enhancement in arc stability is primarily attributed to the increased  $D_{LA}$ , which reduces the impact of metal vapor on the droplets and strengthens the coupling effect between the laser and the arc. When the  $D_{LA}$  is 2–3 mm, the characteristic points are more concentrated, and the welding process is highly stable, with almost no discrete characteristic points. The fluctuation in current and voltage is also minimal, indicating a very stable welding process at this spacing. However, when the  $D_{LA}$  further increases to 4–5 mm (Fig. 4(e, f)), numerous scattered characteristic points appear. The coupling effect between the laser and arc gradually weakens, leading to a reduction in the stability of the welding process. Thus, when the  $D_{LA}$  is 2–3 mm, the coupling efficiency between the laser and arc can be concluded to be the highest, leading to the most stable welding process.



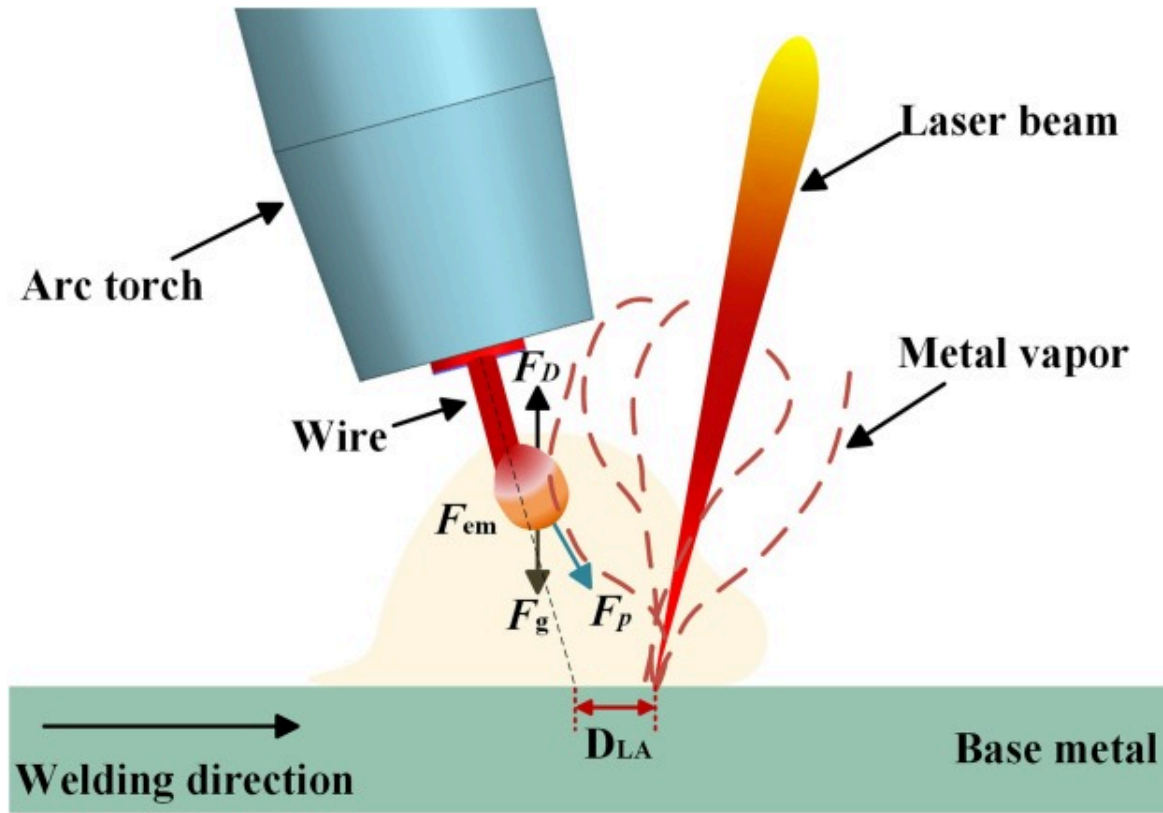
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Fig. 4. Comparison of cyclograms of different laser-arc distances.

### 3.2. Effects of laser-arc distance on droplet force

Based on the analysis in [Section 3.1](#), the  $D_{LA}$  significantly impacts arc stability. The primary factor influencing arc stability is the consistency of droplet transition. In the LAHW process depicted in [Fig. 5](#), according to the static equilibrium theory, droplets within the arc space experience various forces: gravity ( $F_g$ ), plasma flow force ( $F_p$ ), surface tension ( $F_s$ ), electromagnetic force ( $F_{em}$ ), and the reactive force of the metal vapors induced by the laser on the droplet ( $F_D$ ). The droplet enters the molten pool due to the resultant force [\[40\]](#).



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Fig. 5. Schematic of the forces acting on the laser-arc hybrid molten droplet.

When welding horizontally, gravity facilitates the detachment of molten droplets from the end of the welding wire. The gravitational force acting on the droplet can be expressed by [41].

$$F_g = \left\{ \frac{4}{3} \pi r_d^3 - \frac{1}{3} \pi \left[ r_d - (r_d^2 - r_w^2)^{\frac{1}{2}} \right]^2 \left( 2r_d + (r_d^2 - r_w^2)^{\frac{1}{2}} \right) \right\} \rho g \quad (6)$$

where  $\rho$  represents the density of the molten metal of the welding wire,  $g$  denotes the

acceleration due to gravity ( $g=9.8\text{m/s}^2$ ),  $r_d$  represents the radius of the molten droplet, and  $r_w$  indicates the radius of the welding wire.

The plasma flow force is introduced by the airflow from the high-temperature region at the end of the welding wire, which flows rapidly toward the weld pool. This force promotes the separation of molten droplets from the welding wire and influences their trajectory within the arc column region. The plasma flow force  $F_p$  can be expressed as [42].

$$F_p = C_p A \frac{\rho_g v_g^2}{2} \quad (7)$$

where  $C_p$  is the metal vapor drag coefficient ( $C_p=0.45$ ),  $A$  stands for the effective area of the plasma flow (approximated as the cross-sectional area of the molten droplet),  $\rho_g$  denotes the density of the arc plasma flow, and  $V_g$  signifies the velocity of the arc plasma flow. In the flat welding position, when the molten droplet does not touch the weld pool, surface tension  $F_s$  acts as a force that hinders the movement of the molten droplet. Its magnitude can be represented by Eq. (8). When the molten droplet contacts the weld pool, surface tension facilitates its transition. Due to surface tension, the molten droplet will be drawn into the weld pool [43].

$$F_s = 2\pi r_w \gamma \quad (8)$$

$$\gamma_{\text{liq}} = 1.275 - 7 \times (T - 1450) \times 10^{-4} \quad (9)$$

In Eq. (8),  $r_w$  represents the radius of the welding wire, and  $\gamma$  is the surface tension coefficient, the relationship between  $\gamma$  and temperature can be obtained by Eq. (9), where  $T$  is the temperature of the molten pool [44].

The electromagnetic pinch force is mainly generated by the interaction between the current of the molten droplet and its magnetic field. The electromagnetic pinch force can be represented by Eq. (10) [45].

As the electromagnetic force is a vector, it possesses both magnitude and direction. Sometimes it promotes the transition of molten droplets, while at other times it inhibits the transition. According to Eq. (10), the magnitude of the electromagnetic force is directly proportional to the square of the current. Owing to the high viscosity of nickel-based alloys, the transition mode of molten droplets manifests as a rotational jet transition, with the current flowing through the droplet exhibiting a downward expanding distribution. In this case, the downward component of the electromagnetic force is calculated by Eq. (11), which causes the molten droplet to detach from the welding wire.

$$F_{em} = \frac{\mu_0 I^2}{4\pi} F_{emz} \quad (10)$$

The electromagnetic pinch force along the z-direction can be expressed as

$$F_{emz} = - \left[ \frac{1}{4} - \ln \left( \frac{r_d \sin \theta}{r_w} \right) + \frac{1}{1 - \cos \theta} - \frac{2}{(1 - \cos \theta)^2} \ln \left( \frac{2}{1 + \cos \theta} \right) \right] \quad (11)$$

where  $I$  represents the welding current,  $r_d$  stands for the radius of the molten droplet,  $r_w$  denotes the radius of the welding wire,  $\theta$  signifies the arc angle, and  $\mu_0$  is the permeability of free space. ( $4\pi \times 10^{-7} \text{ Hm}^{-1}$ ) [46].

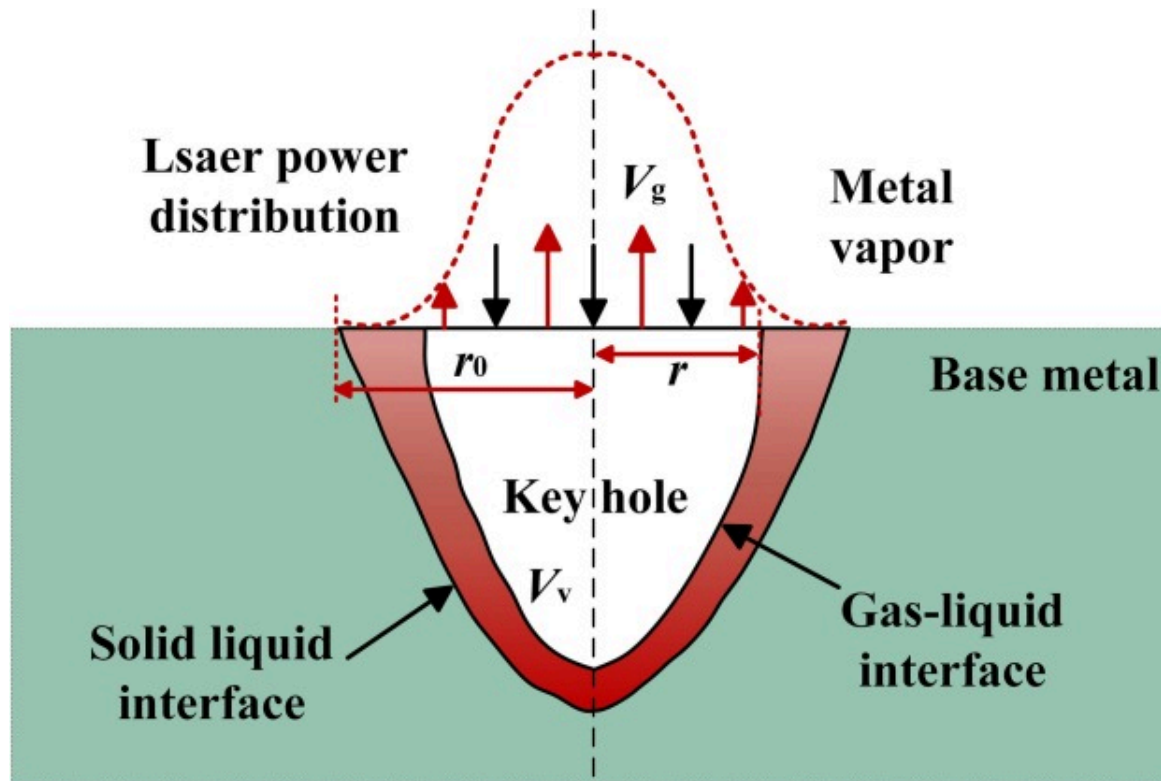
According to the static equilibrium theory, gravity, plasma hydrodynamic force, and electromagnetic force all act as forces promoting the transition of molten droplets. The metal vapor reaction force and surface tension act as resistances. When the driving force acting on the molten droplet exceeds the resistance, and the conditions of Eq. (12) are satisfied, the molten droplet detaches from the tip of the welding wire and enters the molten pool.

$$F_{em} + F_g + F_P > F_S + F_D \quad (12)$$

Due to the high viscosity of the C-276 welding wire, the main transition mode of molten droplets is through rotational jet transition. During the rotational jet process, the liquid jet exhibits radial compression instability. To improve the spiral instability of the liquid

jet, external factors influencing the liquid jet must be strictly controlled. With changes in the  $D_{LA}$ , as indicated by Eq. (12), the main variable is the reaction force of the metal vapor. Therefore, it is necessary to investigate the magnitude and range of the metal vapor reaction force produced by the laser during the LAHW process.

During the LAHW process, the high-energy laser beam interacting with the metal workpiece will cause the welding material to undergo intense evaporation. When the evaporated metal vapor leaves the material surface, it exerts a recoil pressure in the opposite direction on the surface of the molten pool, forming a deep and narrow cavity known as a keyhole. When the recoil pressure of the metal vapor and the surface tension of the liquid metal are balanced, the keyhole can be stabilized [47]. Fig. 6 depicts a schematic of a stable keyhole.



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Fig. 6. Schematic of a stable laser keyhole.

Assuming energy is uniformly distributed within a circular spot with a radius of  $R$ , the effective incident radius of the laser beam energy corresponds to the focal spot radius. This study assumes that the metal vapor generated from the laser keyhole behaves as an incompressible fluid. Following gas dynamics theory, the evaporation of molten metal and the flow of metal vapors are expected to adhere to the mass flow conservation equation:

$$\rho_m v_v = \rho_g v_g \quad (13)$$

In the equation,  $\rho_m$  represents the density of liquid metal, approximated as the density of the C276 welding wire (8.89g/cm<sup>3</sup>), while  $\rho_g$  stands for the density of metal vapors, and  $V_v$  denotes the rate at which the front wall of the molten metal evaporates, according to the Semak model [48]. The surface temperature of the molten metal directly affects the evaporation rate of the front wall in the liquid phase region. Its magnitude can be calculated using Eq. (14), where  $V_v$  is the jet velocity of the metal vapors.

$$v_v = V_0 \exp(-U/T_s) \quad (14)$$

In this equation,  $V_0$  is a constant approximately equal to the speed of sound in the condensed phase (3.4×10<sup>2</sup>m/s),  $T_s$  is the surface temperature of the molten pool, and  $U$  can be calculated using Eq. (15).

$$U = \frac{M_a \cdot L_v}{(N_a K_b)} \quad (15)$$

In Eq. (15),  $M_a$  denotes the atomic mass of the metal vapor (58.69g/mol),  $L_v$  represents the latent heat of evaporation of the molten pool (340.7kJ/kg),  $N_a=6.02 \times 10^{-23}$ , and  $k_B$  is the Boltzmann constant (1.38×10<sup>-23</sup> m<sup>2</sup>kg/s<sup>2</sup>K).

For the molten droplets in the arc space, the impact of the reactive force of metal vapors can be understood as the flow resistance experienced in moving fluids. The magnitude of this resistance depends on the relative movement between the object and the fluid [49]. In a scenario of uniform flow velocity, the reactive force of metal vapors acting on the molten droplets can be represented by Stokes' theory:

$$F_D = C_D A \frac{\rho_g v_g^2}{2} \quad (16)$$

$C_D$  represents the flow resistance coefficient associated with the Reynolds number of the metal vapors ( $C_D=0.45$ ), while  $A$  signifies the projected area of the object in the flow direction. It is posited that the metal vapors can be regarded as an ideal gas [50].

Following the Clapeyron equation for ideal gases, the pressure of saturated metal vapors can be articulated as

$$P_s = \rho_g R_0 T_s = \frac{N_a k_B}{M_a} \rho_g T_s \quad (17)$$

In Eq. (17),  $R_0$  is the ideal gas constant. Using Eq. (17), the correlation between the density of metal vapors  $\rho_g$  and the surface temperature  $T_s$  and pressure  $P_s$  of the molten body can be derived, which can be expressed as Eq. (18):

$$\rho_g = \left( \frac{M_a}{N_a k_B} \right) \frac{P_s}{T_s} \quad (18)$$

As the energy increases, the laser energy absorbed by the molten metal surface instigates the vaporization of the molten metal. The faster the flow velocity of the metal vapors, the more potent the reactive force exerted on the molten droplets. The saturated vapor pressure ( $P_s$ ) is dependent on the surface temperature of the molten pool. As the temperature of the molten body rises due to laser irradiation, the activity and evaporation potential of molecules elevate, consequently boosting the saturated vapor pressure of the molten body. The saturated vapor pressure can be mathematically represented by Eq. (19), according to the Semak model:

$$P_s = A_0 P_s(T_s) = A_0 B_0 T_s^{-1/2} \exp(-U/T_s) \quad (19)$$

The parameter  $U$  in the formula can be calculated using Eq. (15), where  $B_0$  represents the vaporization constant of the molten metal ( $3.9 \times 10^{13} \text{ g} \cdot \text{cm}^{-1} \cdot \text{s}^{-2}$ ). The coefficient  $A$  is directly related to the environmental pressure. According to relevant studies [51], it is approximately 0.55 during vacuum evaporation and can reach up to 1 Pa during evaporation at high environmental pressures. The study also shows that coefficient  $A$  approaches its minimum value of 0.55 under actual environmental pressures.

By combining Eqs. (13), (18), (19), the ejection velocity of metal vapors,  $v_g$ , can be obtained as

$$v_g = \rho_m v_v \left( \frac{N_a k_B}{M_a} \right) \frac{T_s}{P_s} = \rho_m V_0 \left( \frac{N_a k_B}{M_a B_0 A_0} \right) T_s^{\frac{3}{2}} \quad (20)$$

Substituting the metal vapor jet velocity  $v_g$  obtained from Eq. (20) and the metal vapor density  $\rho_g$  obtained from Eq. (18) into Eqs. (16), (21) can be derived.

$$F_D = \frac{1}{2} C_D A \rho_m^2 V_0^2 \frac{N_a k_B T_s^{3/2}}{M_a B_0 A_0} \exp(-U/T_s) \quad (21)$$

In Eq. (21), the surface temperature  $T_s$  of the molten pool determines the magnitude of the metal vapor reactive force. Due to the numerous factors influencing welding parameters during LAHW, including laser power, laser beam diameter, welding speed, arc current and voltage, the relationship between welding parameters and melt pool surface

temperature is highly nonlinear and complex, rendering it difficult to obtain through empirical formulas. Therefore, this study experimentally measures the temperature of the molten pool in the LAHW process to obtain the surface temperature  $T_s$  of the molten pool, which is then substituted into Eq. (21) to calculate the reactive force of the metal vapors on the weld droplet.

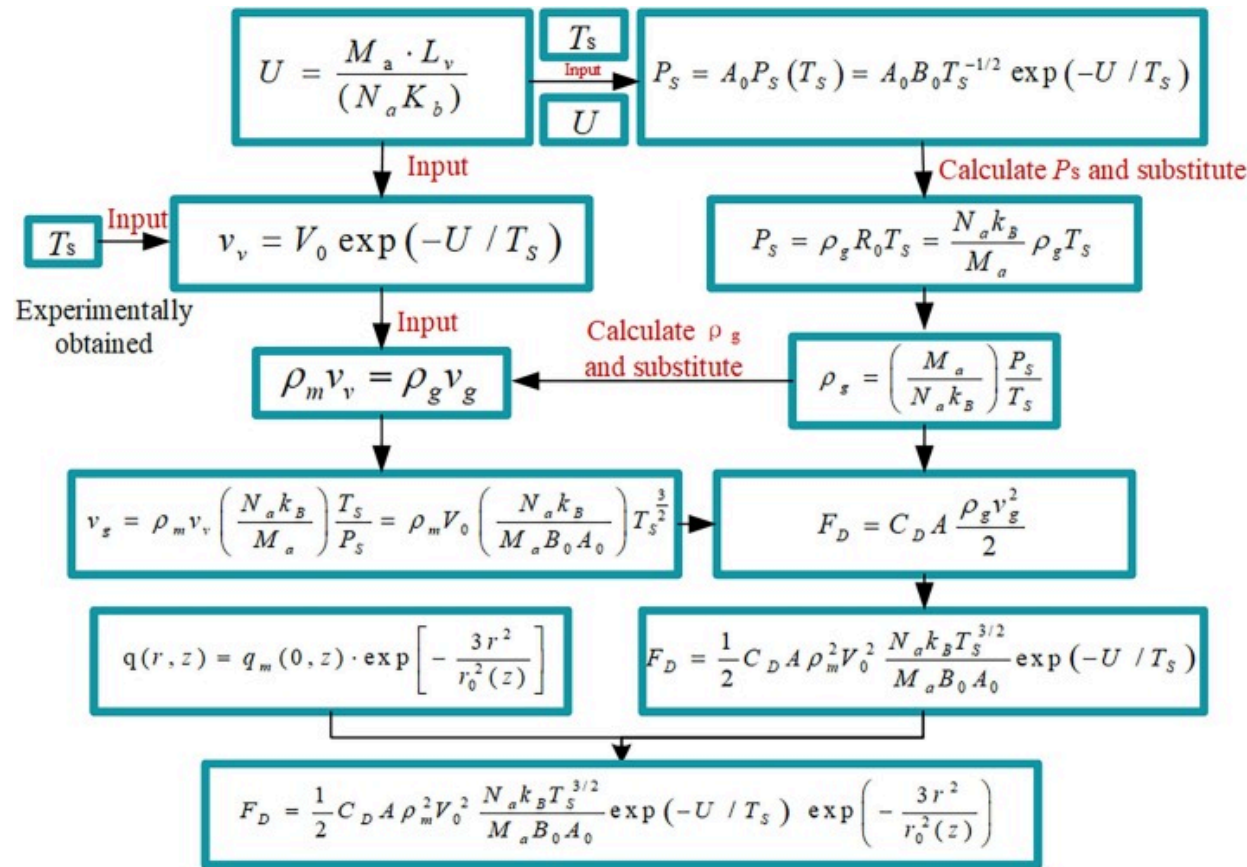
Assuming the laser heat source acts as a Gaussian rotational heat source, and the reactive force of the metal vapors on the weld droplet and its distribution range follow a Gaussian distribution, we have Eq. (22):

$$q(r, z) = q_m(0, z) \cdot \exp\left[-\frac{3r^2}{r_0^2(z)}\right] \quad (22)$$

The radius of the heat flow is denoted as  $r_0(z)$ . In the thickness direction of the workpiece, the radius of the heat flow area exhibits a systematic diminution. The distribution of the metal vapors produced during laser welding is delineated by Eq. (23):

$$F_D = \frac{1}{2} C_D A \rho_m^2 V_0^2 \frac{N_a k_B T_s^{3/2}}{M_a B_0 A_0} \exp(-U/T_s) \exp\left(-\frac{3r^2}{r_0^2(z)}\right) \quad (23)$$

The curve of the recoil force of metal vapors under laser action with the variation of the distance from the heat source can be calculated from Eq. (23). The specific calculation process is shown in the flowchart in Fig. 7. In Eq. (23),  $T_s$  represents the temperature of the molten pool in the laser action area. Due to the highly complex processes of rapid melting and solidification of metals involved in laser action, significant errors exist in calculations using relevant analytical models. Therefore, in this study, experimental data correction of the molten pool in the laser action area was conducted using an infrared thermometer.  $T_s$  was measured experimentally by recording the temperature of the molten pool. The measured temperature of the molten pool was 1900K under a laser power of 3kW using the welding parameters of the experiment. The corrected  $T_s$  was substituted into Eq. (23) and calculated using MATLAB, resulting in the curve shown in Fig. 7.



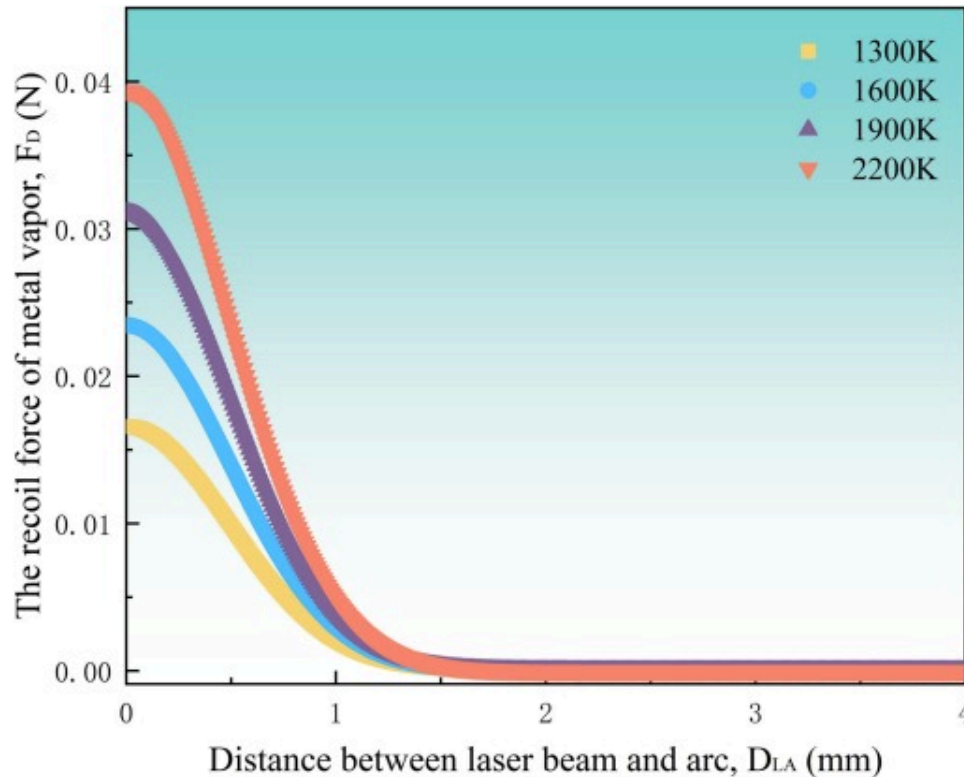
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Fig. 7. The specific calculation process.

Fig. 8 shows that as the temperature of the molten pool within the laser action zone rises, the recoil force of the metal vapors also increases in tandem with the augmentation of laser power. At a molten pool temperature of 1900K, the recoil force of the metal vapors measures 0.03N. When the  $D_{LA}$  is 0mm, the recoil force of the metal vapors is at its maximum, which significantly affects the stability of the arc and droplet transition. With a gradual increment in  $D_{LA}$ , the recoil force of the metal vapors decreases. At a  $D_{LA}$  of 1.5mm, the recoil force of the metal vapors approaches zero. At this point, the impact of

the metal vapors generated by the laser keyhole on droplet transition is minimal, validating the rationale for the compromised stability of the arc at 0mm  $D_{LA}$  as discussed in [Section 3.1](#).



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Fig. 8. Relationship between  $D_{LA}$  and metal vapor recoil force.

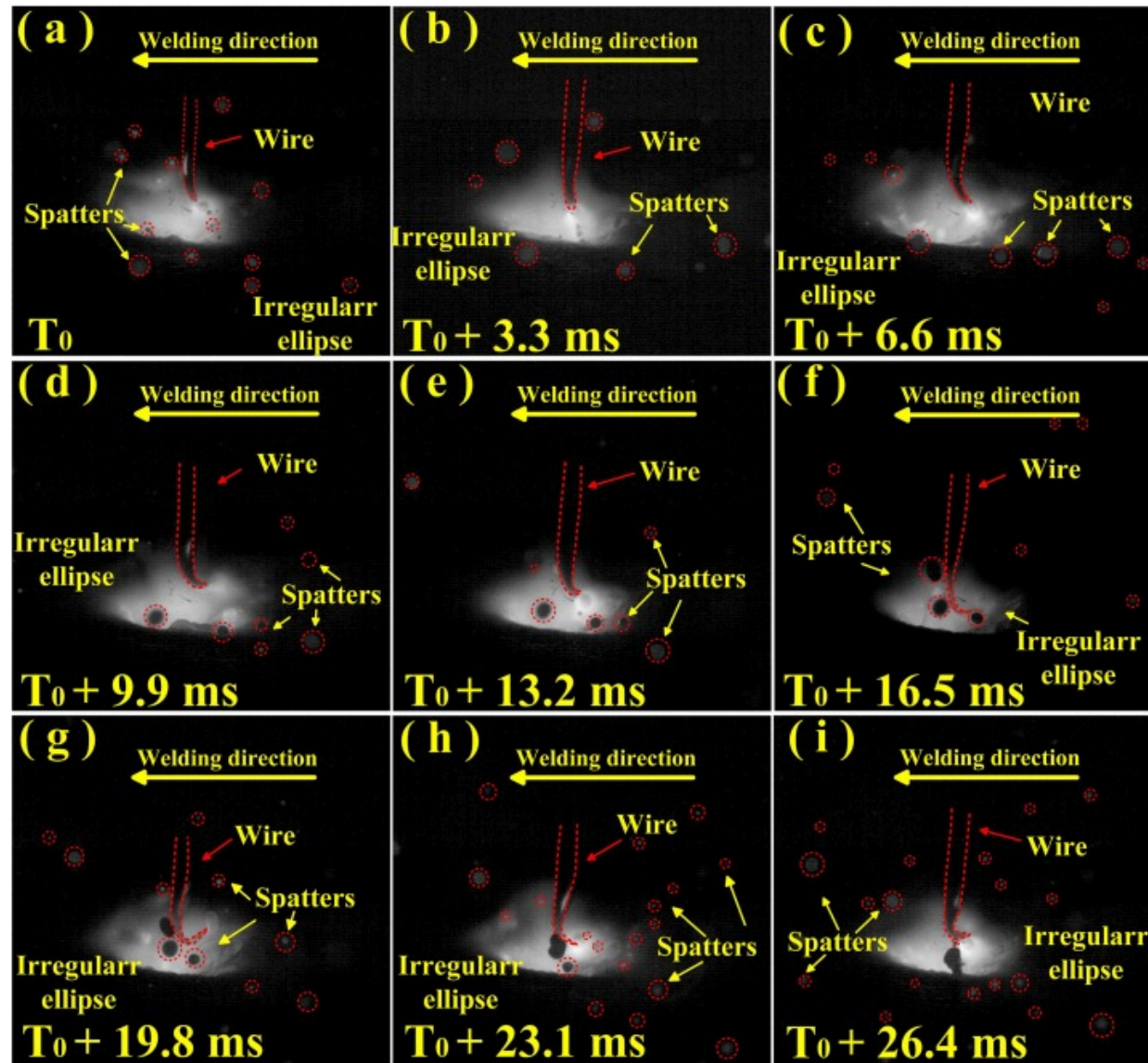
### 3.3. The influence of laser coupling effects on the stability of droplet transition

[Fig. 9](#) depicts images captured by high-speed photography in the absence of laser action. The transition form of the molten droplet is primarily characterized by irregularly oscillating jet transitions. The tip of the welding wire presents a conical shape, and the area where the droplet contacts the solid has softened. Consequently, the droplet finds it



difficult to detach from the wire and enter the molten pool. This phenomenon is primarily attributed to two reasons: first, the high surface tension of elements such as Ni, Mo, W, and Ti in the welding wire makes it challenging for the droplet to fall off the wire into the molten pool; second, the change in the position of the arc root occurs after the wire softens. Compared to when the arc root envelops the entire droplet, in this case, the arc root occupies only a small portion below the droplet, as illustrated in Fig. 9(c). This alteration in the arc root position concentrates the anode spots at the tip of the droplet. At this moment, the current density at the anode spots is high, resulting in concentrated heat generation, which causes excessive heating of the metal and generates a large amount of metal vapors. The metal is ejected perpendicularly onto the droplet surface, producing a reactive force against the droplet and hindering the particle formation of the molten metal. Simultaneously, the tip of the welding wire exhibits irregular oscillation under the reactive force of the metal vapors, as depicted in Fig. 9(g). When the droplet at the tip of the wire is pushed upward by the arc force, it becomes difficult to detach. Additionally, the slender liquid column formed by the softened wire is located quite close to the molten pool. At this point, the slender liquid column is heated to high temperatures by resistive heating, leading to a sudden breakup, and at the moment the droplet detaches, according to the principle of minimum voltage, the arc tends to form between the tip of the liquid column and the molten pool, resulting in significant spattering when the large droplet contacts the molten pool. Moreover, the direction of the recoil force produced by the breakup determines the oscillation direction of the welding wire, causing it to oscillate irregularly. This behavior occurs because the energy flux entering the cathode is generated in a thin region of the near-cathode plasma, and elements such as Ni, Mo, W, and Ti in the C276 welding wire are all difficult-to-evaporate metals with high effective escape work, which makes it challenging to generate sufficient plasma to stabilize and attract the arc. During the welding process, the electron escape work of the oxide film on the surface of the molten pool is small, and the cathode spots of the arc always tend to seek aggregation of oxide points on the micro-surface of the metal. However, oxides of Ni, Mo, W, etc., remain unstable at high temperatures, making it difficult to provide stable oxides to constrain the oscillation of the arc. Based on the above

analysis, this paper intends to employ laser-arc hybrid welding to improve welding efficiency while introducing a stable conductive pathway through laser action. This will regulate the coupling effect between the laser and the arc plasma.



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### Fig. 9. High-speed image of the welding process without laser.

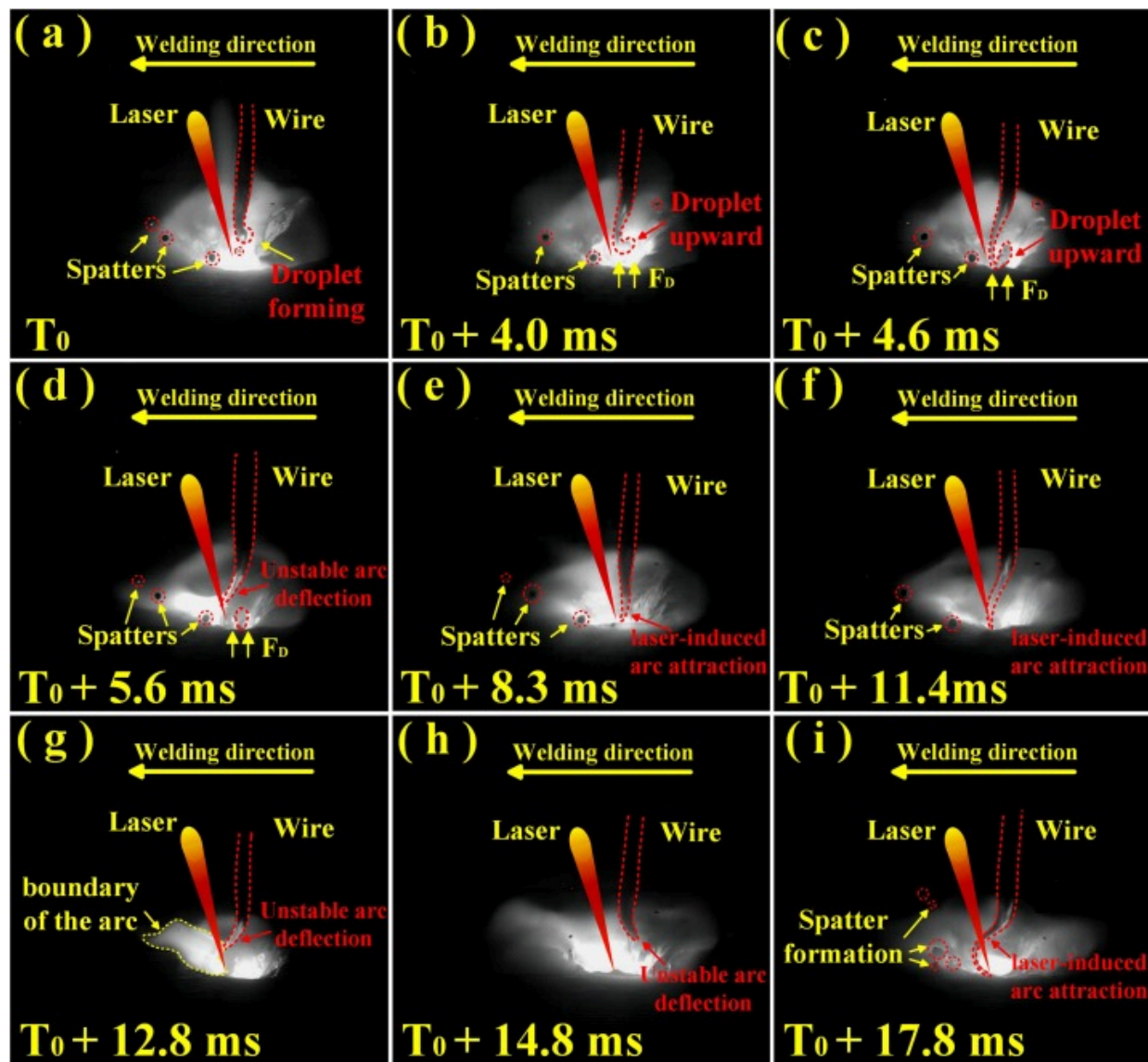
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Based on the stability analysis of the LAHW process under different  $D_{LA}$  in [Section 3.1](#), an optimal solution for enhancing the stability of the welding process emerges. Moreover, [Section 3.2](#) shows the range and magnitude of the reactive force of metal vapors generated by the laser during the LAHW. Further analysis of high-speed images of the processes with  $D_{LA}$  of 0mm to 5mm showcase how the coupling effect between the laser and the arc influences the stability of the process.

Owing to the high viscosity of nickel-based alloys and the large surface tension of the melted wire tip, the progression of molten droplets is more challenging compared to materials such as low-alloy steel, making the transition process more complex. Additionally, there is limited research on LAHW of nickel-based alloys.

[Fig. 10](#) shows images captured by high-speed cameras with a  $D_{LA}$  of 0mm. At  $T_0$ , molten droplets gradually take shape at the wire tip under the combined influence of the laser and the arc plasma ([Fig. 10\(a\)](#)). At  $T_0+4.0\text{ms}$ , with a  $D_{LA}$  of 0mm, based on the analysis in [2 Materials and experimental procedure](#), [3 Results and discussion](#), the molten droplet is impacted by the counteracting force of the laser metal vapors, causing the end of the liquid column to exhibit a tendency to incline upwards. By  $T_0+4.6\text{ms}$ , the metal liquid column deviates from the center of the arc and is repelled upward. The molten droplet, under the influence of gravity and electromagnetic force, is unable to counteract the surface tension and the opposing metal vapor force, resulting in the formation of a neck constriction. Consequently, when the softened portion of the wire tip column comes into contact with the tip of the molten pool, significant splattering transpires, with the splashes moving upward, and a small number of molten droplets not transferring to the molten pool. After the molten droplet completely detaches, at  $T_0+5.6\text{ms}$ , due to the higher local electron density on the side where the laser is acting, a conductive channel with high conductivity is provided, causing the arc to swing toward the laser side. By  $T_0+8.3\text{ms}$ , due to the counteracting force of the metal vapors from the laser affecting the liquid column, the arc plasma is not stably attracted by the laser and moves away from

the laser toward the rear side of the molten pool. At  $T_0 + 12.8\text{ms}$ , the wire liquid column continues to be periodically attracted by the laser plasma. [Fig. 11](#) illustrates how the transitional behavior of molten droplets is influenced when the  $D_{LA}$  is 0mm. The liquid column rotational oscillation around the laser is accompanied by a significant amount of upward splashing under the influence of the laser.

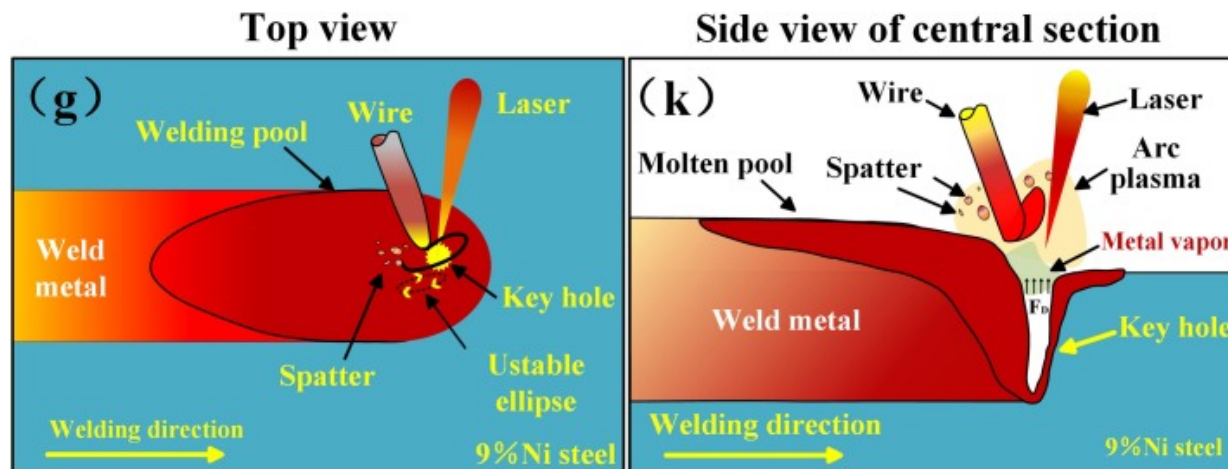


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Fig. 10. High-speed image of the welding process with a  $D_{LA}$  of 0mm.





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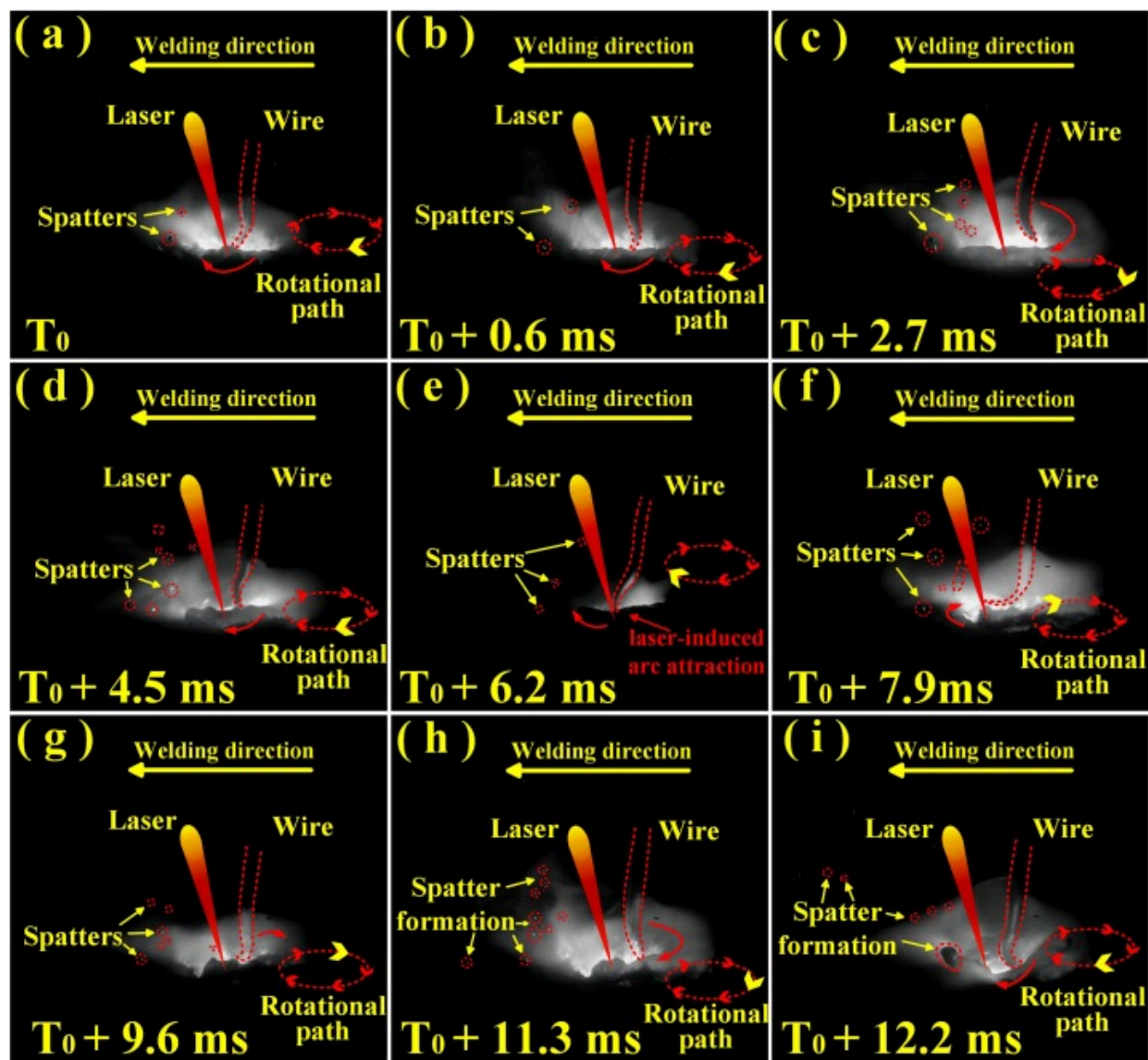
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Fig. 11. Schematic of molten droplet transition behavior with a  $D_{LA}$  of 0mm.

Fig. 12 shows images captured by high-speed cameras with a  $D_{LA}$  of 1 mm. Throughout the entire welding process, there is no obvious formation of molten droplets at the wire tip, and the diameter of the conical liquid column uniformly varies along the axial direction. The soft and curved portion of the wire tip has a certain rigidity when rotating. Compared to a  $D_{LA}$  of 0mm, the rotating radius of the liquid column at the wire tip increases while the rotation speed also increases. Due to the weak wetting and spreading capabilities and high viscosity of C276 alloys, the arc rotates within a smaller radius, approximately 1.5mm, making a circular motion. The arc is unable to effectively preheat the base material, which is not conducive to the wetting and spreading of C-276 on the base material. Under the action of the arc force, the center of the molten pool sinks, showing a characteristic of being higher around the edges and lower in the middle. The arc is periodically attracted and repelled by the laser. Following the principle of minimum voltage, the raised edges of the molten pool offer a better conductive route for the arc, facilitating periodic rotation of the arc around these raised edges. At  $T_0+7.9\text{ms}$ , as the arc revolves within the laser's range, it draws nearer to the laser source. Under the

counteracting force of the metal vapors produced by the laser, the arc undergoes an upward shift, leading to a significant amount of upward splashing, [Fig. 13](#) illustrates how the transition behavior of droplets is affected with a wire spacing of 1 mm. While the welding process's stability is notably enhanced compared to a 0mm wire spacing, there are still unstable phenomena when the rotating wire column reaches the laser-affected area.

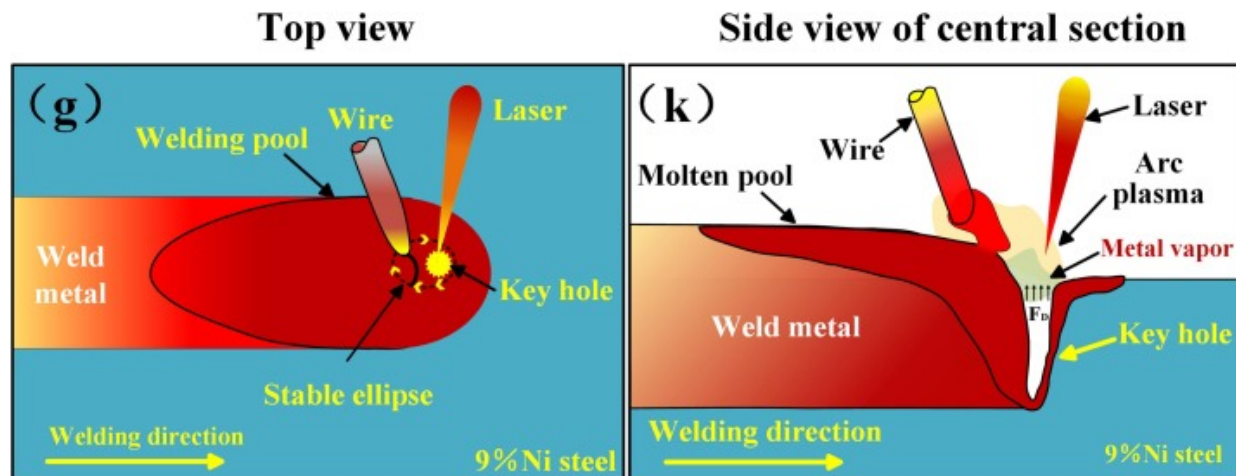




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Fig. 12. High-speed image of the welding process with a  $D_{LA}$  of 1 mm.



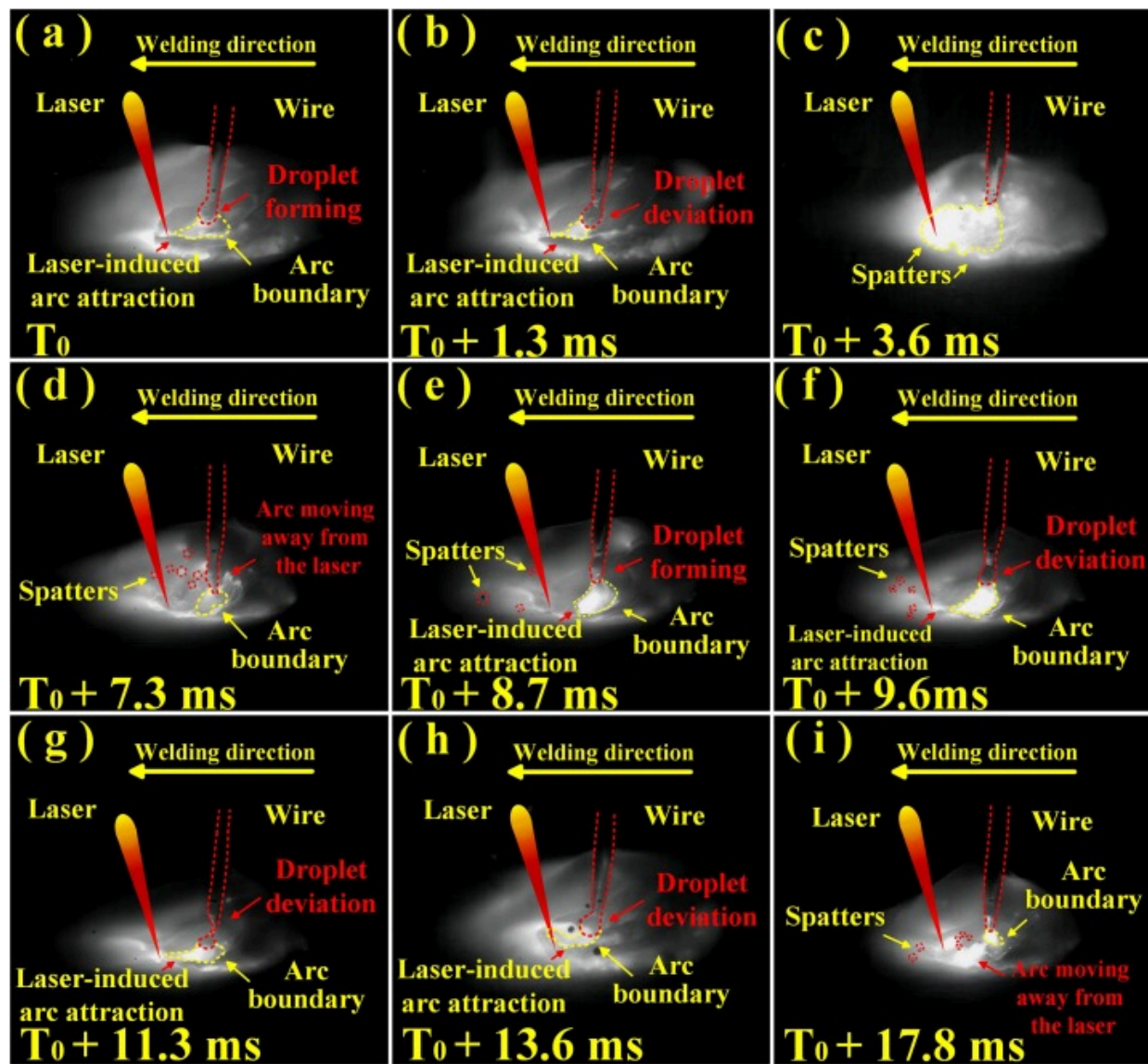
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Fig. 13. Schematic of molten droplet transition behavior with a  $D_{LA}$  of 1 mm.

Fig. 14 shows images captured by high-speed cameras with a  $D_{LA}$  of 2 mm. Compared to the rotational spray transition exhibited with  $D_{LA}$  of 0 mm and 1 mm, the molten droplet transition state at this point shows a free droplet transfer. At  $T_0$ , the droplet forms at the welding wire tip, and the arc is deflected toward the laser side under the influence of the laser plasma. At  $T_0 + 1.3$  ms, the droplet gradually enlarges, and under the influence of electromagnetic force and plasma flow force, it is deflected toward the laser side. By  $T_0 + 3.6$  ms, the droplet detaches from the tip of the welding wire and transitions toward the end of the molten pool. Due to the concentrated laser energy and small effective area, the wetting and spreading of the molten pool primarily rely on the arc heating the molten pool metal along the direction of the laser action. At this point, with the  $D_{LA}$  being only 2 mm, the heated area is small after the arc deflection, compounded by the high viscosity and poor wetting and spreading capabilities of nickel-based alloys. This causes the droplet to fall at the front of the molten pool. As the droplet contacts the molten pool, it instantaneously breaks apart, leading to forward spattering. At the moment of droplet ejection, the arc extinguishes. At  $T_0 + 7.3$  ms, after re-ignition, due to the short distance

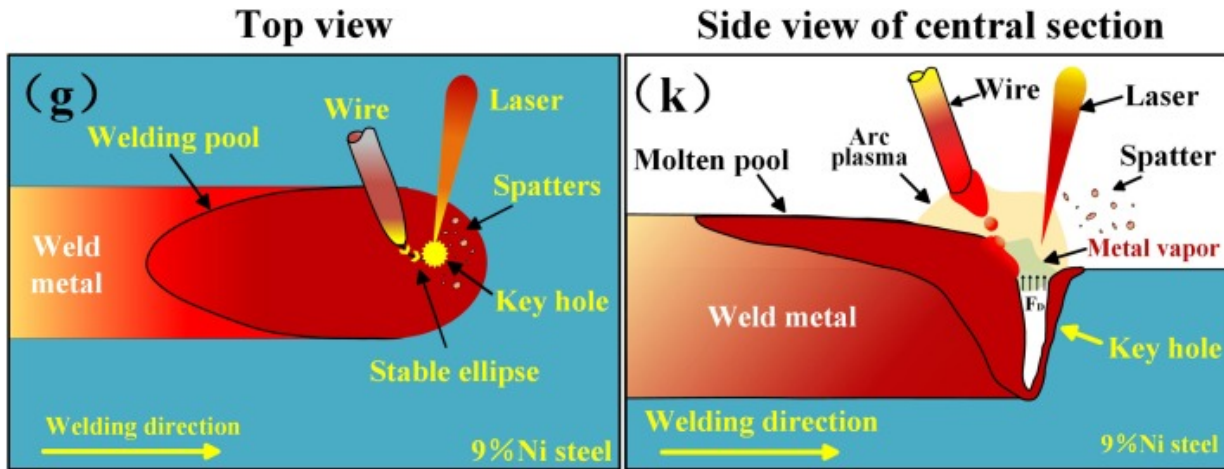
between the wire tip and rear of the molten pool, based on the minimum voltage principle, the arc re-ignites in a position away from the laser at  $T_0+8.7\text{ms}$ . From  $T_0+9.6\text{ms}$  to  $T_0+11.3\text{ms}$ , it gradually moves toward the direction of the laser plasma. [Fig. 15](#) illustrates how the transition behavior of droplets is affected with a wire spacing of 2 mm. Under the influence of the laser, the liquid column shows periodic deflection toward the laser to complete the droplet transition, leading to spattering and moving away from the laser.



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Fig. 14. High-speed image of the welding process with a  $D_{LA}$  of 2 mm.



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Fig. 15. Schematic of molten droplet transition behavior with a  $D_{LA}$  of 2 mm.

Fig. 16 shows images captured by high-speed cameras with a  $D_{LA}$  of 3 mm. When the  $D_{LA}$  is 3 mm, the transition mode of the molten droplet exhibits a projected transition. At  $T_0$ , the molten droplet gradually forms at the wire tip. At this time, the laser is far from the arc compared to the laser distance, and the molten droplet is not influenced by the reactive force of the laser metal vapors. At  $T_0 + 0.7$  ms, the arc deflects due to the influence of the laser plasma, and simultaneously, the molten droplet begins to deviate from the wire tip under the influence of electromagnetic forces. Compared to a  $D_{LA}$  of 2 mm, the increased  $D_{LA}$  results in the arc being attracted and deflected by the laser over a larger range, providing sufficient preheating of the metal at the front of the molten pool. The relationship between metal viscosity and temperature based on the Andrade–Eyring formula can be expressed by Eq. (24) [52].

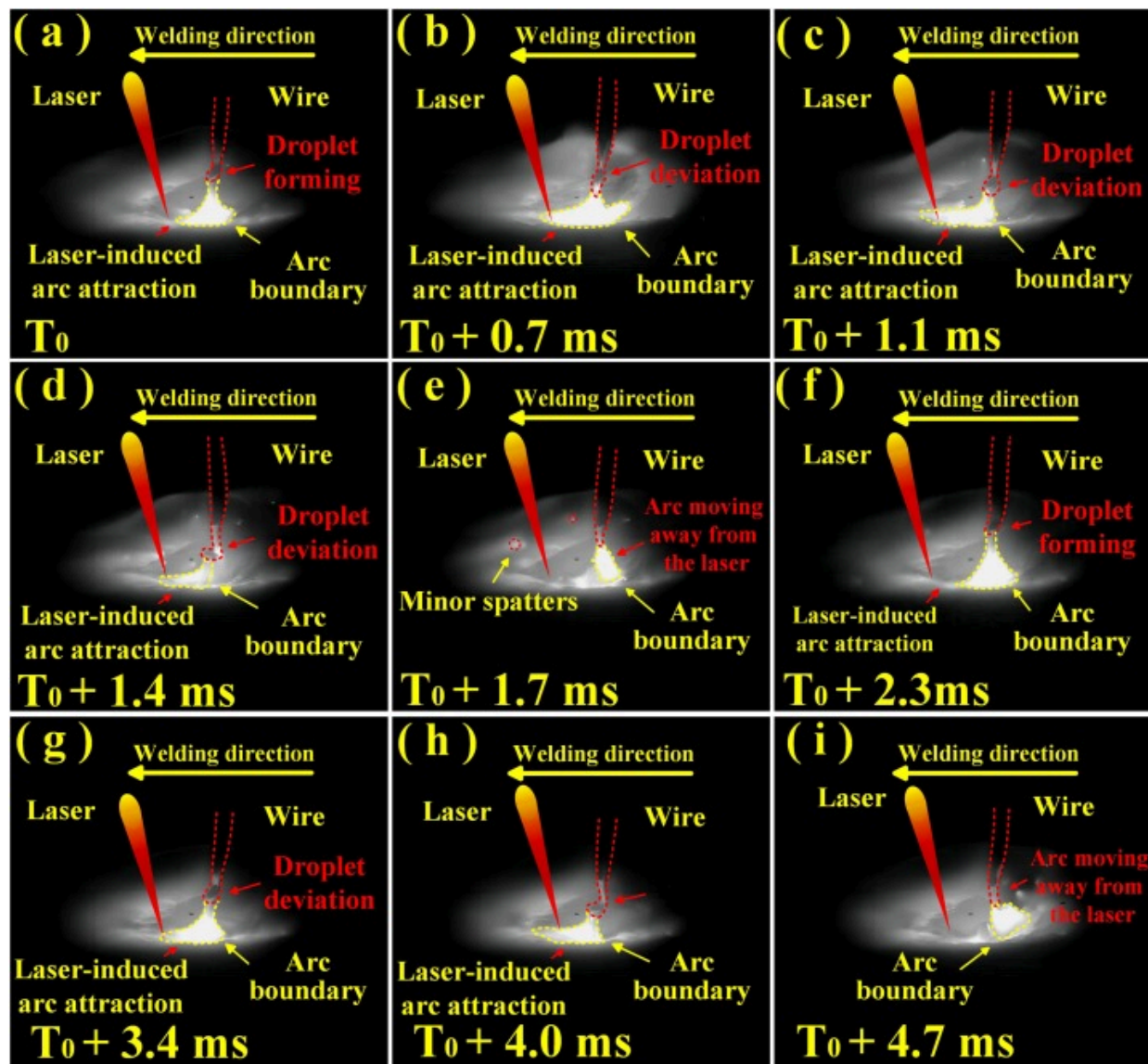
$$\eta = A * \exp(B/T) \quad (24)$$

where  $\eta$  represents the viscosity of the metal, and A and B are constants correlated to the metal properties, while T represents temperature. Eq. (24) illustrates that as temperature



rises, the viscosity of molten metal diminishes, which increases the fluidity of the molten pool. At  $T_0 + 1.4\text{ms}$ , the molten droplet separates from welding wire tip and moves toward the side closer to the laser. Based on the increased fluidity of the molten pool, the droplet enters the molten pool with almost no splashing or only a small amount of splashing. At  $T_0 + 1.7\text{ms}$ , after detaching from the tip of the welding wire, the molten droplet swings away from the laser, with the main reason being that at this time, the molten droplet has not yet formed at the tip of the welding wire, and the distance between the welding wire and the laser plasma is far. By  $T_0 + 2.3\text{ms}$ , the molten droplet gradually forms at the tip of the welding wire, and the liquid column at the tip of the wire softens. At this point, the arc continues to be attracted by the laser plasma. Fig. 17 illustrates a schematic of the trajectory of the molten droplet in the electrical space with a wire spacing of 3mm. The arc periodically deviates toward the laser, and the droplet moves away from the laser after detachment.

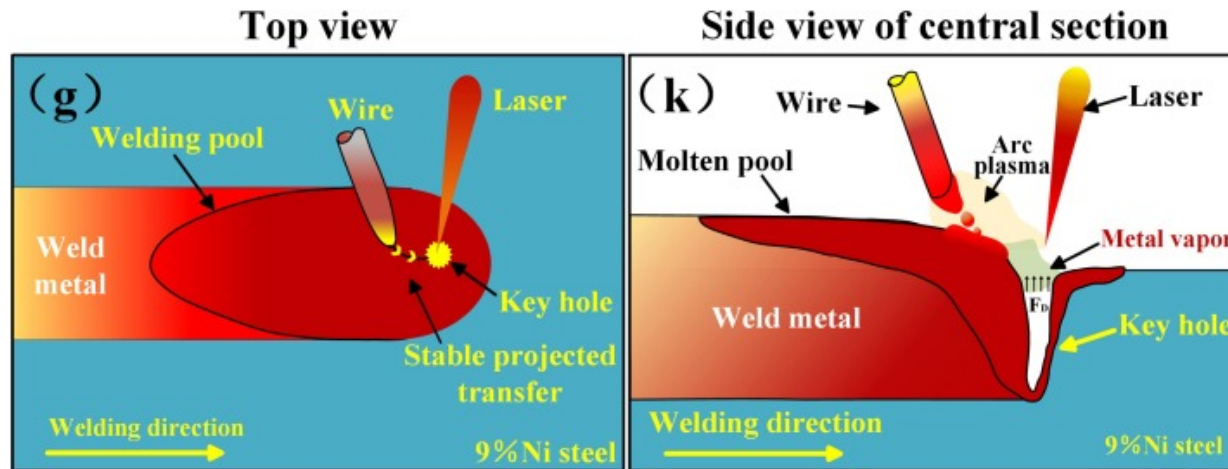




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Fig. 16. High-speed image of the welding process with a  $D_{LA}$  of 3 mm.

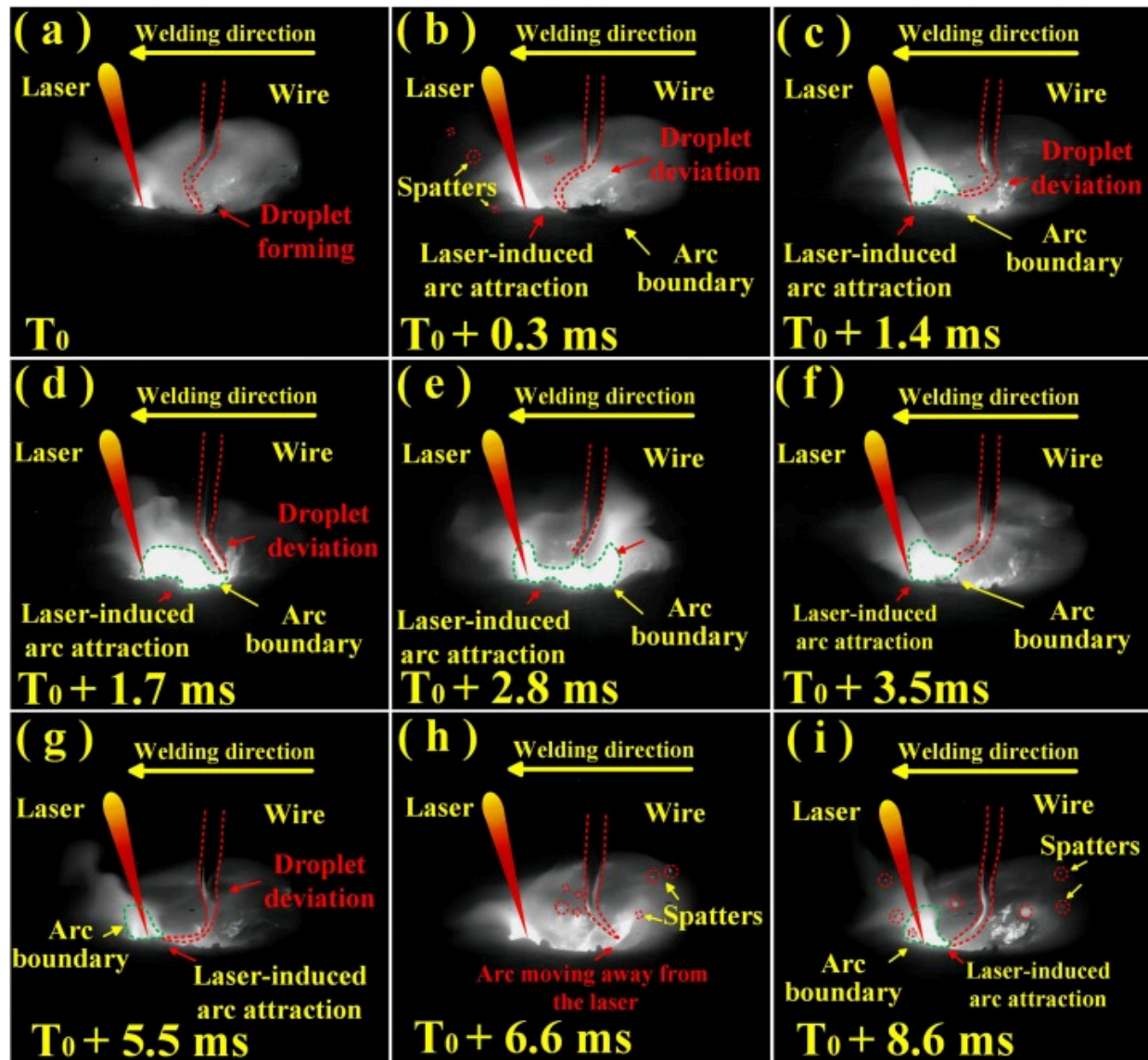


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Fig. 17. Schematic of molten droplet transition behavior with a  $D_{LA}$  of 3 mm.

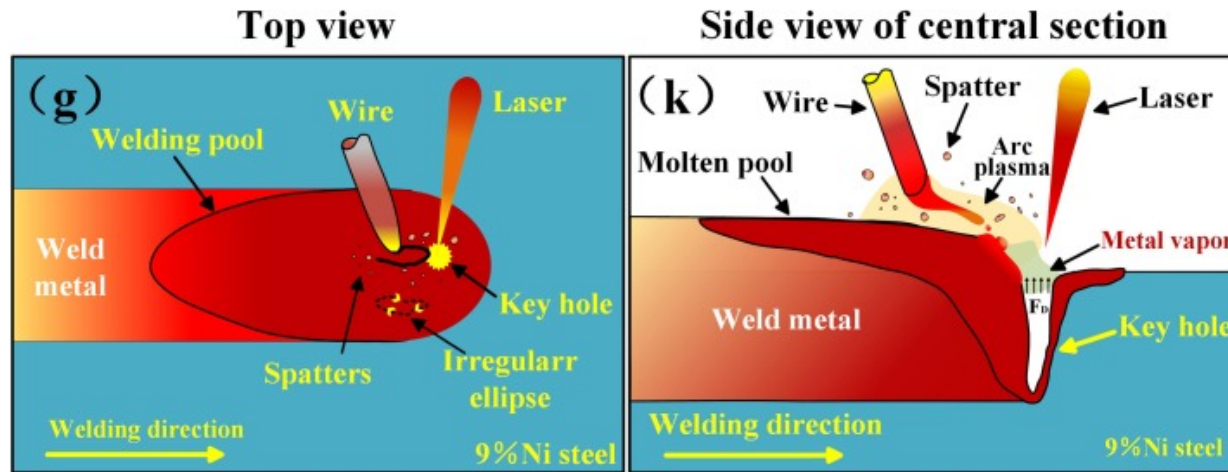
Fig. 18 shows images captured by high-speed cameras with a  $D_{LA}$  of 4 mm. When the  $D_{LA}$  is 4 mm, the molten droplet transition takes the form of an irregular rotating spray transition. At  $T_0$ , the wire tip softens, and at  $T_0+0.3$  ms, the liquid column at the wire tip deflects toward the laser. Simultaneously, at  $T_0+1.4$  ms, the arc is drawn toward the laser plasma. However, as the laser is far from the arc at this point, the laser plasma cannot attract the arc for an extended period. At  $T_0+1.7$  ms, after the molten droplet transition is completed, the arc deflects away from the laser, resulting in significant spatter during the arc deflection transition process. Fig. 19 illustrates the schematic trajectory of molten droplet movement in the electrical space with a  $D_{LA}$  of 4 mm, where the liquid column at the wire tip exhibits irregular rotation.



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Fig. 18. High-speed image of the welding process with a  $D_{LA}$  of 4mm.



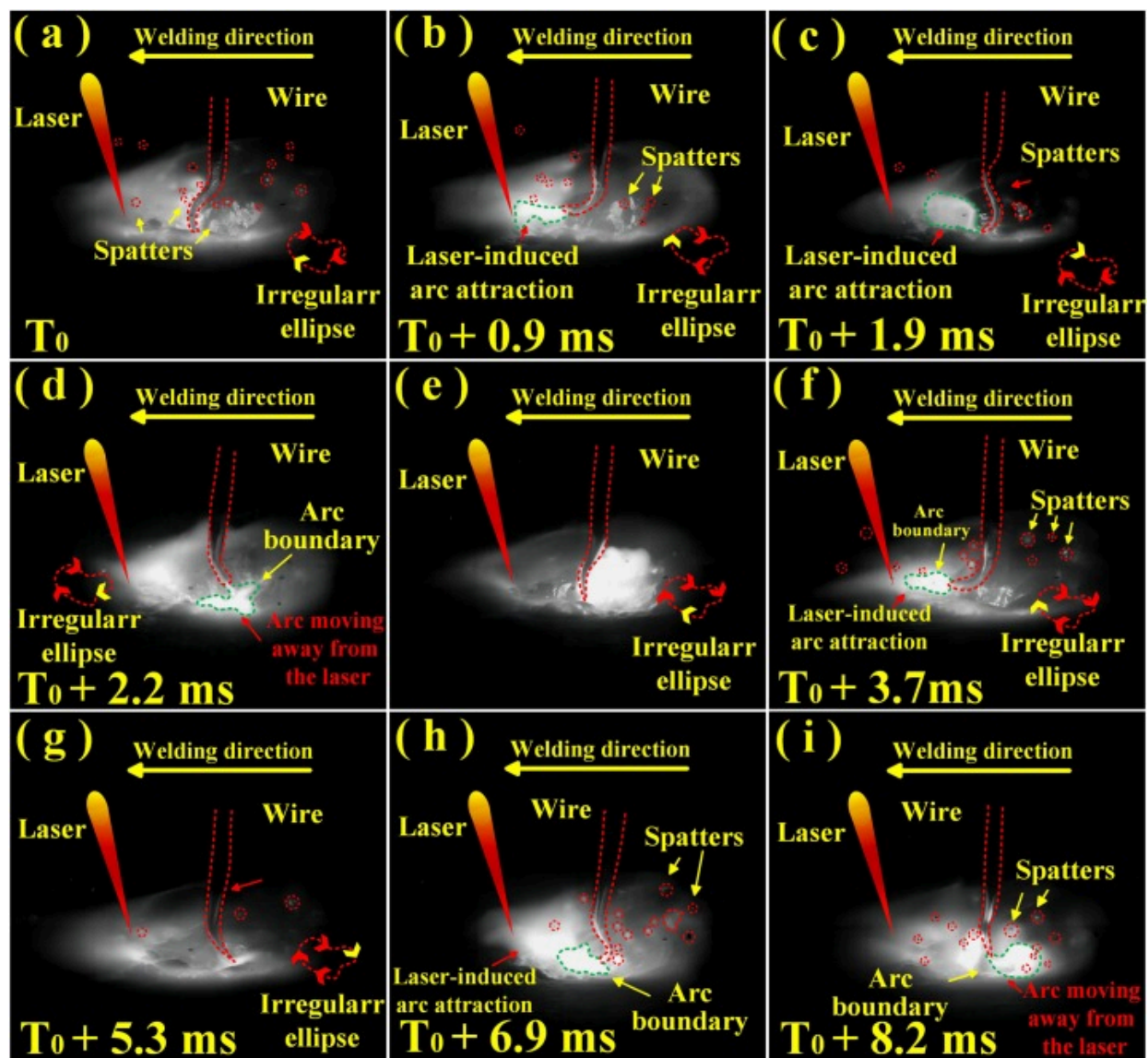
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Fig. 19. Schematic of molten droplet transition behavior with a  $D_{LA}$  of 4mm.

Fig. 20 shows images captured by high-speed cameras with a  $D_{LA}$  of 5mm. At a  $D_{LA}$  of 5mm, the molten droplet transition takes the form of an irregular rotating spray transition. At  $T_0$ , the wire tip softens, and at  $T_0+0.9ms$ , the liquid column at the wire tip deflects toward the laser. Since the laser is relatively distant from the arc at this juncture, the arc deflection is irregular. During this irregular deflection process, the liquid column contacts the molten pool, leading to wire fragmentation at the interface surface between the liquid column and the molten pool, resulting in significant spattering, as depicted in Fig. 20(h). Fig. 21 illustrates the schematic trajectory of molten droplet movement in the electrical space with a  $D_{LA}$  of 5mm, where the liquid column at the wire tip exhibits irregular rotation accompanied by a significant amount of spatter.

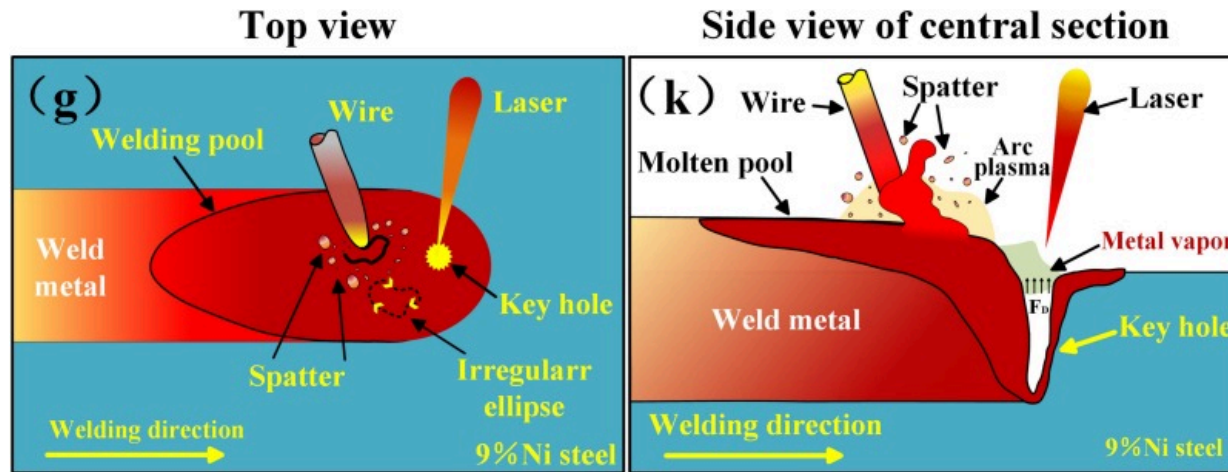




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Fig. 20. High-speed image of the welding process with a  $D_{LA}$  of 5 mm.



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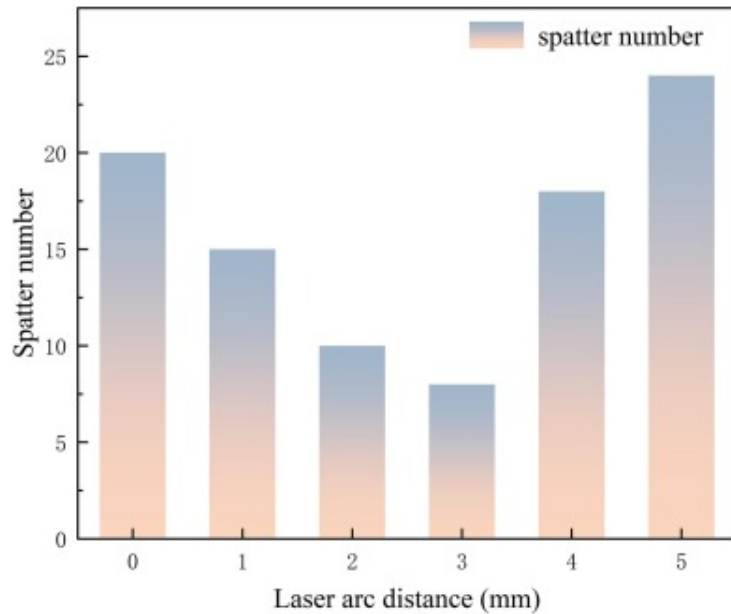
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Fig. 21. Schematic of molten droplet transition behavior with a  $D_{LA}$  of 5 mm.

To further investigate the influence of different laser-arc distances on the stability of the laser-arc hybrid welding process, high-speed imaging technology was employed to visually record and statistically analyze the spatter behavior during welding. Under various laser-arc distance conditions (ranging from 0 mm to 5 mm), while keeping all other welding parameters constant, frame-by-frame analysis of high-speed video sequences was conducted to quantify the number of spatter particles generated per droplet transfer event. This approach enabled the evaluation of process stability and the effectiveness of spatter suppression. As shown in Fig. 22, the number of spatter particles significantly decreased as the laser-arc distance increased from 0 mm to 3 mm, indicating improved welding stability. However, when the distance further increased to 4–5 mm, the spatter count rose again, suggesting a weakening of the coupling effect between the laser and the arc, intensified molten pool fluctuations, and reduced process stability. These results further confirm that maintaining a laser-arc distance within the range of 2–3 mm yields optimal welding performance under the given process conditions. In this interval, the weld penetration is maximized, fluctuations in welding current and voltage are



minimized, and spatter generation is significantly suppressed, thereby enhancing both process stability and thermal efficiency, and contributing to the formation of high-quality welds.



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Fig. 22. Statistical analysis of spatter number per droplet transfer under different wire spacing conditions.

## 4. Conclusions

This study examined the impact of laser on the stability of the transfer process of nickel-based alloys by analyzing the coefficients of variation and cycle graphs of welding current and arc voltage at different  $D_{LA}$ . It found that the addition of laser can effectively enhance the stability of the transfer of high-viscosity metals. Additionally, the  $D_{LA}$  significantly affects the stability of the welding process. As the  $D_{LA}$  increases, the stability of the welding process initially improves and then deteriorates.

A mathematical analytical model for the metal vapor recoil force generated by the laser in the LAHW process was established. Temperature parameters were experimentally corrected, and the metal vapor recoil force acting on the droplet was quantitatively calculated. The range of action of the metal vapor recoil force under the laser was also determined. The results indicate that when the  $D_{LA}$  exceeds 1 mm, the influence of the metal vapor recoil force generated by the laser on the molten droplet is minimal.

By observing the droplet transition behavior under different  $D_{LA}$  through high-speed camera images, the study found that at a DLA of 0mm, the droplet was significantly influenced by the metal vapor recoil force from the laser. At wire spacing of 0mm, the liquid column oscillates around the laser spot. At a wire spacing of 1 mm, the droplet transition exhibited a small-radius spray transition. When the wire spacing is between 2 and 3 mm, the welding process is most stable. As the wire spacing continues to increase, the coupling effect between the laser and the arc weakens, and the droplet transfer behavior exhibits irregular rotational spray transitions.

## CRediT authorship contribution statement

**Zhenwen Zhu:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Yu Shi:** Conceptualization. **Gang Zhang:** Investigation. **Chunkai Li:** Project administration. **Ming Zhu:** Formal analysis. **Ziyou Ren:** Software. **Wenkai Wang:** Validation. **Yang Zhai:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

This work was supported by the central government's special funds for guiding local science and technology development (grant number [24ZYQA054](#)), the major science and technology projects-industry (grant number [23ZDGA010](#)), the Project of the Excellent Ph.D. Student Project in Gansu Province (grant number [23JRRA771](#)), Gansu Provincial Key Research and Development Program (Grant no. [25YFGA023](#)), and Innovation Fund for Higher Education Institutions of Gansu Province (Grant no. [2025A-022](#)).

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